

修 士 論 文

**ネットワーク計画ツールとしての欲求経路創発
シミュレーションマルチエージェントシステムの開発と評価**

**Development and Evaluation of a Multi-Agent System
simulating Emergence of Desire Paths as Networks
Planning Tool**

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第1章 Introduction

In recent years, researchers have increasingly turned their attention to nature's ability to solve complex problems, such as optimizing transportation networks. This section explores various models inspired by natural phenomena and behaviors, examining how individual interactions drive the self-organization of systems. Building on these insights, we propose a hybrid multi-agent system designed to create optimized transportation networks, striking a balance between minimizing the total network size and reducing travel time.

The following topics are discussed in this chapter:

- Research background
- Previous works
- Objectives

1.1 Research background

This research focuses on the design of efficient path system that connect critical points, or hotspots, in a way that optimizes both performance and adaptability to real-world constraints. These networks are vital across a wide range of applications, including urban planning, telecommunications, energy distribution, water systems, traffic prediction, neuroscience, autonomous vehicle navigation, and even game development.

By leveraging advanced optimization models, we aim to balance competing objectives such as minimizing construction or operational costs, reducing travel or transmission time, and ensuring robustness against dynamic demands and constraints. Such work provides valuable insights and tools for improving the efficiency and sustainability of complex systems.



Fig. 1.1: Illustration of the ability of a multi-agent system (humans walk) to create paths in environment.

1.2 Previous research

To reproduce real-world transport systems and networks, my research employs a variety of approaches that integrate bio-inspired algorithms, graph theory, and optimization techniques. These methods aim to achieve a balance between minimizing total network size, reducing travel times, and ensuring robustness under real-world constraints. For instance, the application of Steiner trees offers an effective solution for designing efficient path systems by minimizing the overall network length while connecting key nodes.

Researchers aimed their attention to other solutions like bio-inspired multi-agent model and algorithms to solve NP-hard problems, in particular slime mold and pedestrian movement.

This discussion explores the use of these models and algorithms, evaluating their performances and the constraints they encounter in real-world applications.

1.2.1 Spanning and Steiner trees

Steiner trees are a concept in graph theory and network optimization used to connect a set of points (known as terminal nodes) with the shortest possible total network length. Unlike direct spanning trees, which connect all points directly without introducing any new nodes, Steiner trees allow the inclusion of additional points (called Steiner nodes) to reduce the overall network length.

For example, imagine trying to connect cities with roads to minimize construction costs. A Steiner tree might introduce intermediate junctions to achieve a more efficient design, resulting in a shorter total network than if the cities were simply connected directly.

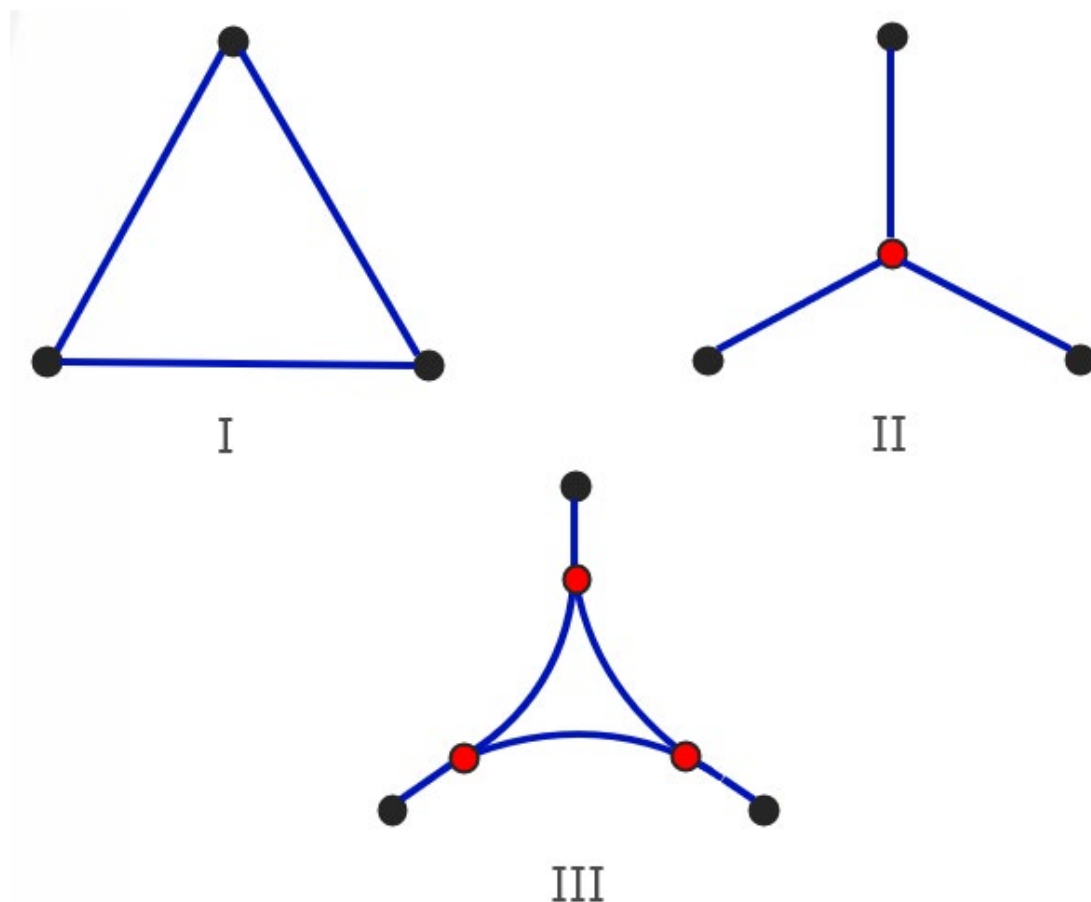


Fig. 1.2: (I) Spanning tree solution with direct connections, (II) Steiner tree and (III) Intermediate solution for three points arranged in a triangular configuration.

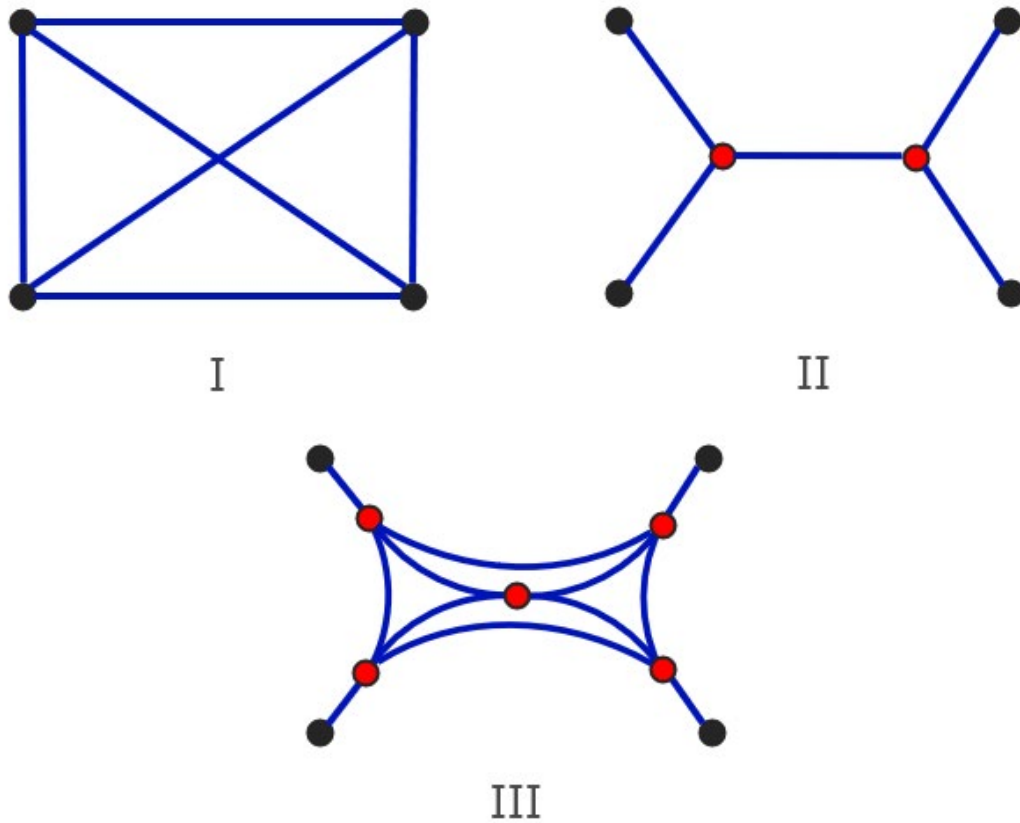


Fig. 1.3: (I) Spanning tree solution with direct connections, (II) Steiner tree and (III) Intermediate solution for four points arranged in a square configuration.

However, Steiner trees are not always ideal for real-world scenarios due to several practical limitations and the mathematical calculation of Steiner trees can be complex and approximated [1]. First, Steiner trees focus solely on minimizing construction or connection costs and do not account for other critical factors such as travel time, reliability, or adaptability to changing demands. In transportation networks, for instance, the shortest possible network might result in indirect routes, leading to increased travel time and reduced practicality for users.

Second, real-world networks often involve additional constraints such as geographic barriers, political boundaries, or infrastructure requirements, which Steiner trees do not inherently consider. For example, a road network optimized purely for minimal length might cross areas where construction is impractical or prohibited, such as mountains, rivers, or private land.

Finally, Steiner trees assume static conditions, whereas real-world systems are dynamic, with fluctuating demand, traffic patterns, or environmental factors. A network designed as a Steiner tree may lack the robustness or flexibility needed to handle disruptions or accommodate growth, making it less suitable for long-term applications.

To address these challenges, more adaptable solutions are necessary—ones that balance the cost-efficiency of Steiner trees with the practicality of spanning trees. These intermediate approaches prioritize not only minimizing network size but also improve travel time, usability, and resilience to real-world constraints, resulting in networks better suited for practical applications like transportation, telecommunications, or urban planning.

Furthermore, a more dynamically adaptable model—capable of self-reconfiguring until it reaches a stable state within a predefined simulated environment—can provide a powerful tool for designing optimized transportation networks. Such a model would allow for the iterative refinement of the network, accounting for factors like fluctuating demand, geographic constraints, and environmental changes, ultimately leading to a more robust and efficient system tailored to real-world conditions.

1.2.2 Slime mold behavior for urban network design

The slime mold model provides valuable insights into adaptive, self-organizing systems that mimic efficient resource distribution [2-7]. Based on the natural behavior of the *Physarum Polycephalum*, it has inspired the development of a multi-agent algorithm designed to emulate its remarkable ability to create efficient transport networks. In this system, each agent is modeled with the same morphology and behavior, mimicking the decentralized and self-organizing characteristics of slime mold. These agents operate independently, following simple local rules to explore and connect resource points within an environment.

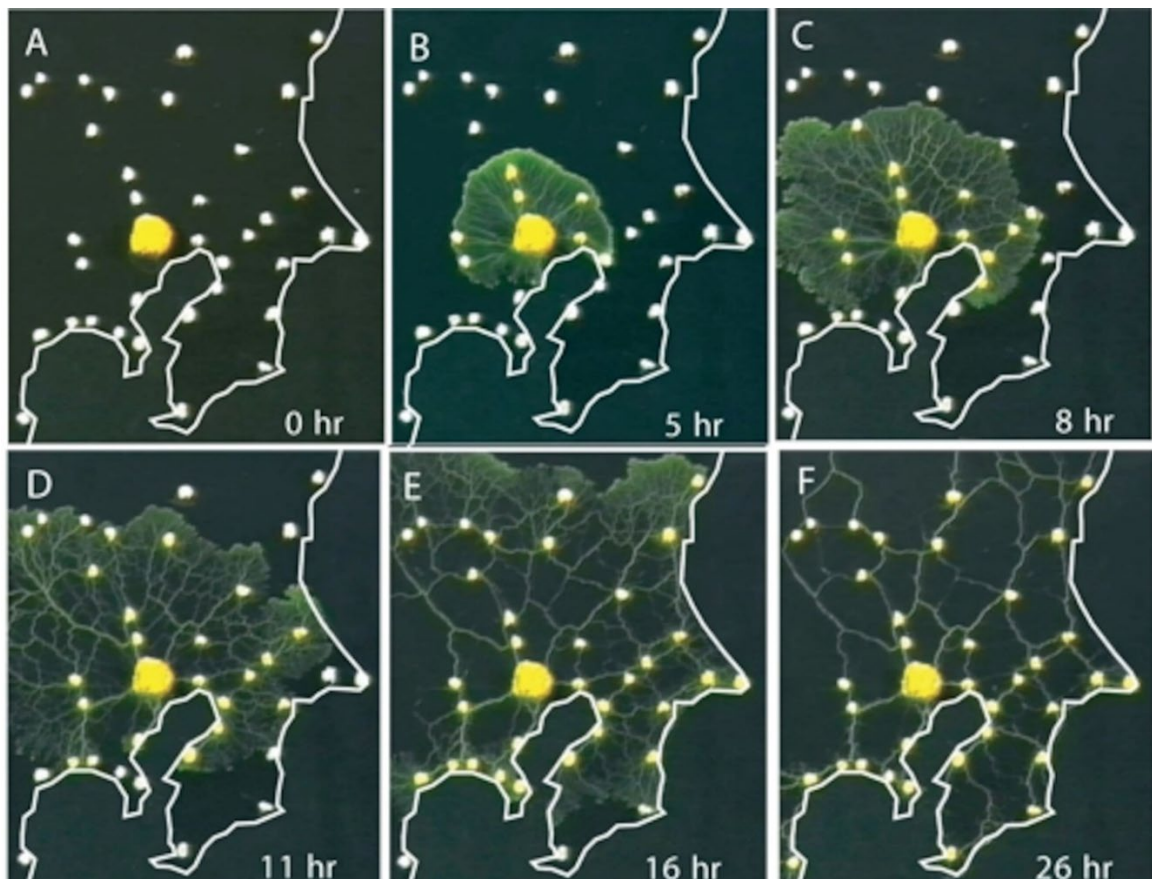


Fig. 1.4: *Physarum Polycephalum*'s ability to recreate the Tokyo railway system: oats representing stations and slime mold growth forming an optimized transport network after 26 hours.

When deployed in large numbers, these agents collectively form intricate networks that optimize path efficiency and resource distribution. Remarkably, this approach mirrors slime mold's ability to recreate real-world transportation systems, such as the Tokyo railway network, by finding near-optimal connections between key nodes while minimizing overall network length. This bio-inspired algorithm demonstrates how emergent behavior in simple agents can lead to sophisticated, scalable solutions for complex urban design challenges.

Each agent is composed of four key components, including three sensors—forward (F), front-right (FR), and front-left (FL)—which detect the concentration of stimuli in the surrounding environment. These sensors guide the agent's movement by influencing its position and orientation. Based on the stimuli detected, the agent either rotates toward the strongest local source of stimuli or continues moving forward in its current direction.

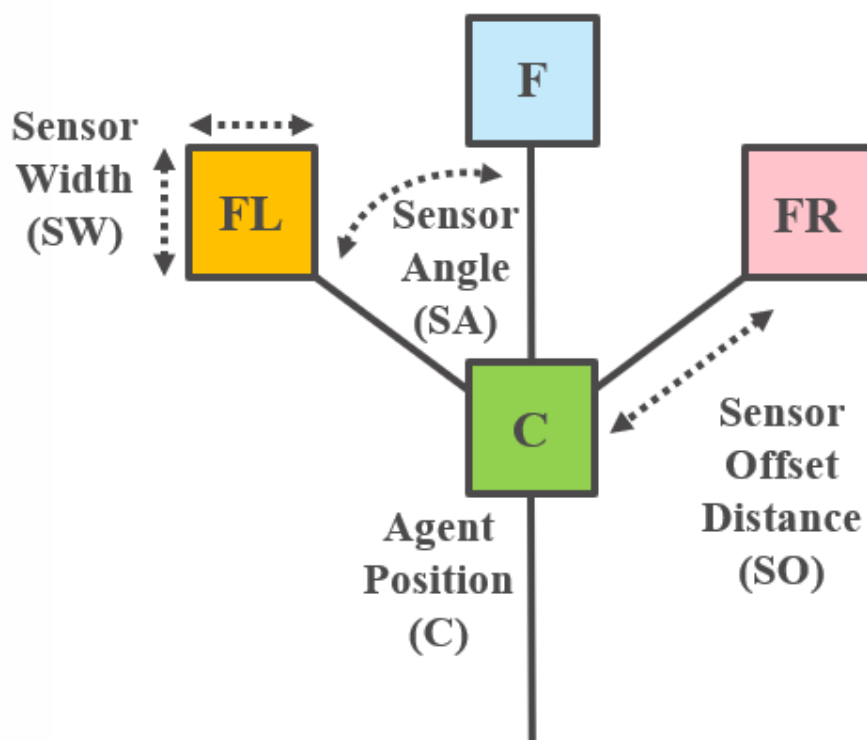


Fig. 1.5: Agent morphology of slime mold model

Table 1.2: Parameters for sensing

SO	Sensor offset	SA	Sensor angle
SW	Sensor width	C	Agent position
FL	Forward left	FR	Forward right
F	Forward		

Figure 1.5 and Table 1.2 illustrate the agent's morphology, showing the arrangement of the sensors and the behavioral parameters that dictate its responses. The behavior of the agents can be fine-tuned by adjusting two critical parameters: the sensor offset (SO) and the sensor angle (SA).

Sensor Offset (SO) determines the spatial distance between the agent and its sensors. Increasing the SO value enhances the agent's foraging behavior, leading to the formation of thicker, more interconnected networks and enabling faster adaptation to environmental changes. Conversely, a smaller SO reduces foraging tendencies, promoting grouping behavior and tighter clustering of agents.

Sensor Angle (SA) defines the angular placement of the sensors relative to the agent's forward direction. Smaller SA values result in agents moving along straighter paths, which may cause them to bypass food sources or fail to group effectively. In contrast, larger SA values allow the agent to better detect stimuli from a wider range of angles, improving its ability to forage and adapt to the environment.

By carefully balancing these parameters, the agents' behaviors can be tailored to simulate various scenarios, from foraging and exploration to grouping and clustering. This flexibility makes the model highly adaptable for studying emergent path systems and their optimization in dynamic environments.

1.2.3 Pedestrian movement to reproduce desire paths

The pedestrian movement model contributes to understand human dynamics within networks, capturing the influence of desire paths and real-world behavioral tendencies [9].

The pedestrian movement behavior implemented in this multi-agent algorithm is designed to replicate the self-organizing phenomenon of desire paths, facilitating the creation of efficient, optimized networks. The model is based on simple yet effective rules inspired by human spatial cognition and movement patterns observed in urban environments. Each agent in the model represents an individual pedestrian, equipped with two primary visual parameters – angle of vision and depth of vision. The interaction between these parameters defines the agent's "viewshed," or the area within which they perceive and respond to environmental cues and affordances.

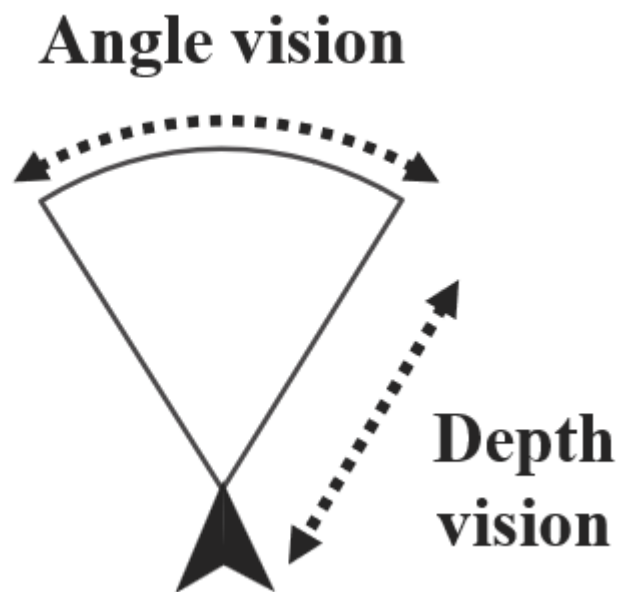


Fig. 1.6: Visual parameters of agent in pedestrian movement model

Desire paths arise from the repeated passage of individuals at specific locations. By deploying a large number of agents, each individual interacts with their environment and modifies it, ultimately leading to the formation of a global network pattern. However, a key feature of this model is its dynamic representation of human movement.

Unlike a simplistic trajectory-following approach, the agents' orientation is not fixed solely to their current path. Instead, they direct their view area toward their destination, regardless of their immediate trajectory. This behavior mimics how humans—and other living beings—often turn their head or change their focus to anticipate and navigate their surroundings, even when their current movement is directed elsewhere. This aspect introduces a higher level of dynamism into the agents' behavior, allowing them to adapt their paths more fluidly in response to both immediate and anticipated environmental factors.

By enabling the agents to orient their attention away from their current trajectory, the model incorporates a more natural, human-like adaptability. It allows for a more nuanced and realistic representation of movement, where agents can adjust their route or make real-time decisions based on new stimuli or emerging traffic patterns and optimize their journey.

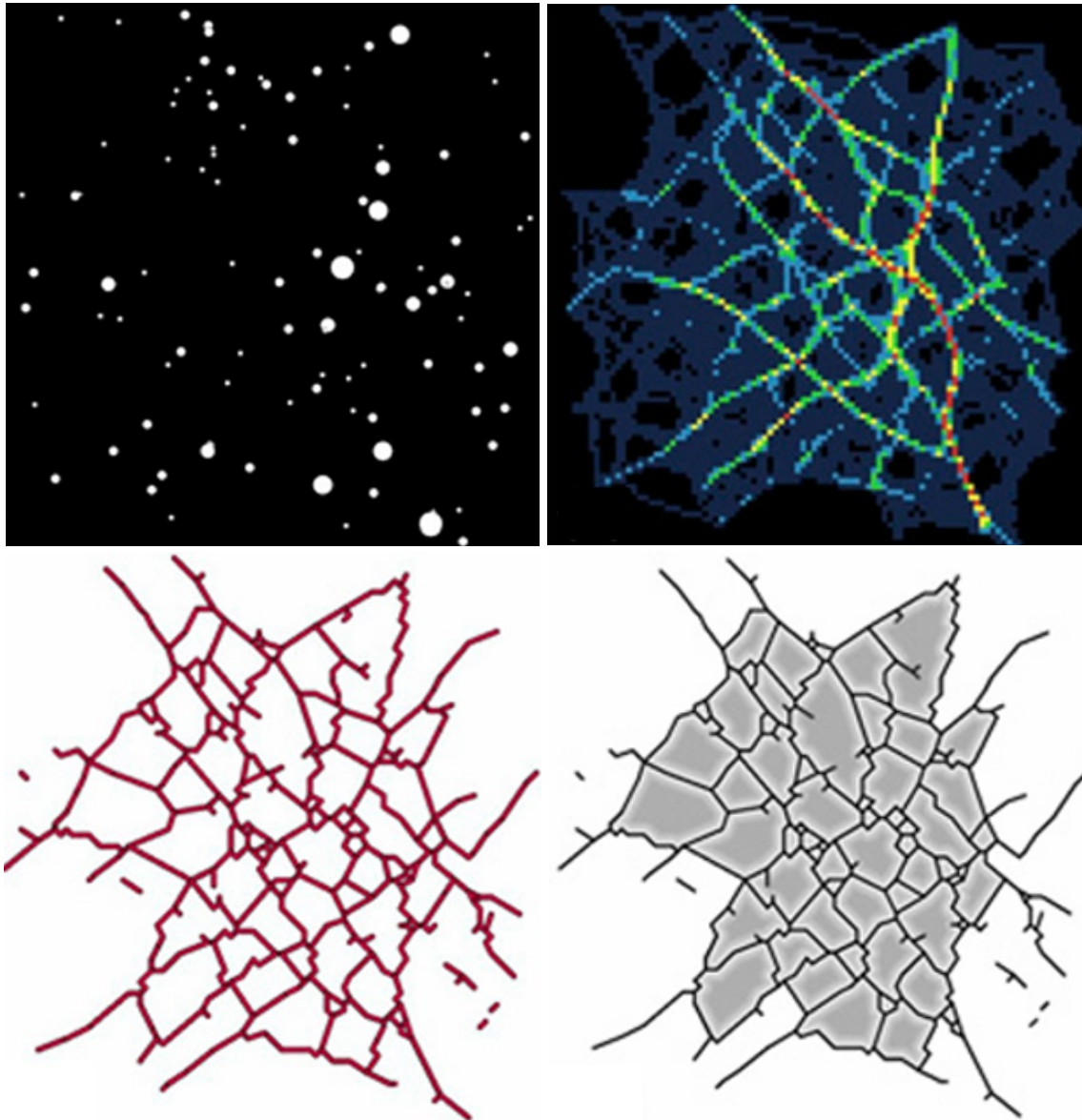


Fig. 1.7: Multi-agent system reproducing pedestrian movement of humans from a heterogeneous distribution of hotspots in terms of visit frequency (up left), resulting in the emergence of a path system (up right) and traffic density of roads. A path system (bottom left), extracted by discarding patches with smaller affordance values and connecting the greater ones; and its corresponding enveloped path blocks (bottom right).

1.2.4 General issues with previous models

While previous models have made significant strides in replicating natural behaviors and optimizing network designs, they often face several limitations. These include a lack of adaptability to dynamic environments, the emergence of inefficiencies such as incomplete coverage or deviated paths, and challenges in balancing competing objectives like minimizing network length and ensuring practical usability. Additionally, while comparisons to real human-designed path systems have been conducted, there has been little to no evaluation of performance in terms of total network size and the ability to account for complete travel counts. This section explores these general issues, highlighting the gaps that need to be addressed to improve the reliability and applicability of such models in real-world scenarios.

1.2.4.1 Slime mold model limitations

If we analyze the agent behavior algorithm, it becomes apparent that agents tend to continue facing the same direction, even in the absence of detectable stimulus concentrations. This behavior can lead to issues such as uncertainty in the direction of the emerging pattern and the formation of incomplete or faulty regions in the resulting network or structure.

Algorithm 1 Agent behavior algorithm

```
[Motor stage]
Attempt move forwards in current direction
if (moved forwards successfully) then
    Deposit trail in new location
else
    Choose random new orientation
end if

[Sensory stage]
- Sample trail map values
if ( $F > FL$  AND  $F > FR$ ) then
    Stay facing same direction return
else if ( $F < FL$  AND  $F < FR$ ) then
    Rotate randomly left or right by RA
else if ( $FL < FR$ ) then
    Rotate right by RA
else if ( $FR < FL$ ) then
    Rotate left by RA
else
    Continue facing same direction
end if
```

This phenomenon occurs because agents, lacking stimuli to guide their movement, rely on their previous orientation, causing them to proceed in a straight or slightly deviated path. While this might simplify local decision-making, it introduces a level of randomness that can disrupt the coherence of the overall system. For example, in a simulated transport network, agents may fail to cover critical areas or may diverge from their intended paths, leading to inefficiencies and gaps in the final design.

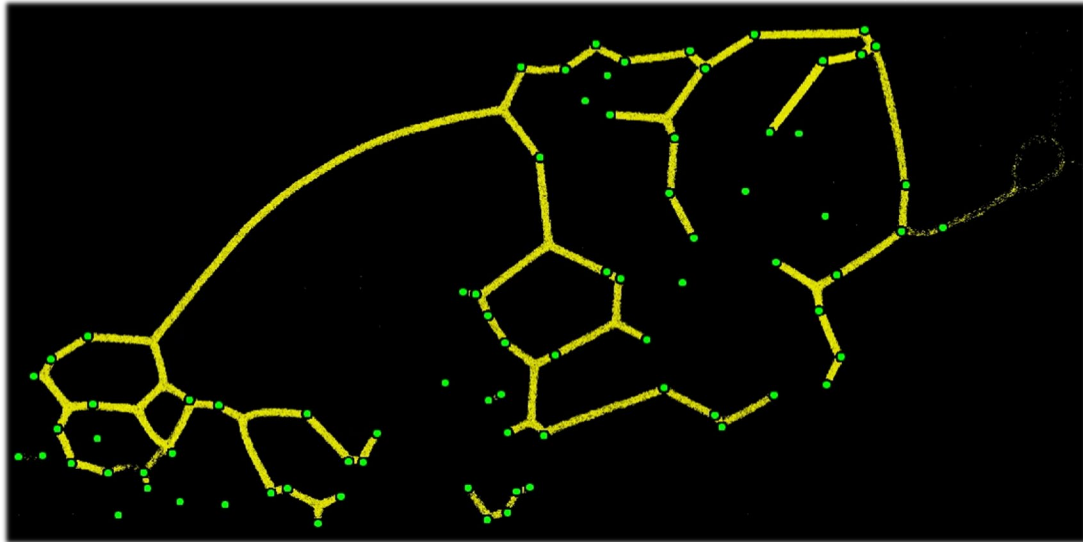


Fig. 1.8: Emergence of Steiner tree patterns, deviated paths and region faults in a multi-agent model simulating slime mold behavior.

Additionally, this behavior highlights the importance of incorporating mechanisms to correct or adapt agent trajectories in the absence of stimuli. Without such mechanisms, the self-organization process can suffer from incomplete coverage or imbalanced resource allocation, which may limit the model's ability to produce efficient, optimized patterns.

1.2.4.2 Desire paths model limitations

In contrast, the pedestrian movement model offers valuable insights by mimicking human behavior, but it falls short when it comes to evaluating the stability of patterns, the performance of the network, or the distribution of road usage frequency. One significant limitation of this approach is its inability to dynamically adapt over time. Despite the emergence of a transport network, unused roads persist in the final output. This static feature hinders the model's capacity to evolve and adapt to changing conditions, as it does not account for the diffusion phenomenon—one where paths dynamically shift based on real-time usage and traffic flow.

The lack of adaptability in the desire paths model is particularly problematic for long-term urban planning and transportation optimization. Human traffic needs are constantly fluctuating, influenced by factors such as time of day, weather conditions, and urban events, which require a system that can evolve in response to these changes. Without the ability to adjust and redistribute traffic flows in real-time, the model is restricted in its effectiveness. For example, if certain roads or routes are underutilized, the system remains static and fails to redirect traffic, which ultimately reduces the overall efficiency of the network.

A more robust and adaptable model would incorporate a time-dependent variable and integrate diffusion processes. By simulating how human traffic needs evolve dynamically, the model could continuously optimize traffic patterns, removing unused roads from active use and rerouting traffic to maintain a balanced, efficient system. This approach would ensure that the resulting network remains flexible, reducing inefficiencies and enhancing performance over time. In applications such as urban planning, traffic optimization, and resource distribution, the model would be capable of adjusting to the ongoing changes in traffic demand, making it more reliable and sustainable for future use.

1.3 Objectives

This research aims to present and evaluate a new multi-agent model capable of creating intermediate solutions while ensuring free-scale adaptability. The model will dynamically adjust to real-time traffic data and demand, balancing key factors such as travel time, total system length, and traffic flow. By incorporating the flexibility to modify routes based on real-time conditions, this approach seeks to enhance the robustness and efficiency of the network. The performance evaluation will focus on assessing the model's effectiveness based on the agents' morphology configuration, examining how different configurations impact the optimization of the network's performance.

第2章 Principles and behavior algorithms

This chapter describes the principles and behavior model of the agent and target selection when performing the transport network system. The following items are discussed in this chapter.

- Destination decision-making principle
- Human visitation law
- Target selection algorithm

2.1 Destination decision-making principle

Choosing a destination is a complex process influenced by a combination of personal preferences, external factors, and contextual circumstances. Whether for leisure, business, or other purposes, individuals and groups rely on various principles to guide their decisions. These principles can be broadly categorized into the following:

- Attractiveness
- Visit frequency

The attractiveness of a place is one of the most significant factors influencing its selection as a travel destination. It encompasses elements such as cultural landmarks, scenic beauty, recreational activities, and the overall ambiance of the location. Highly attractive destinations naturally draw attention and become popular among travelers.

Places with unique characteristics, such as the Tokyo Skytree's breathtaking views or Shinjuku's vibrant nightlife, hold a strong allure for tourists. The attractiveness of a location is often shaped by several key factors. Aesthetic appeal, including beautiful landscapes, architectural marvels, and unique urban designs, plays a significant role in drawing visitors. Cultural significance, embodied in historical landmarks and cultural events, adds depth and meaning to a place, making it more compelling. Additionally, the availability of amenities and activities, such as entertainment, shopping, and diverse dining options, enhances the overall experience for visitors. Together, these elements establish high attractiveness, making a place a "must-visit" destination and encouraging both initial and repeat visits.

However, their frequent visitation and resulting crowd density can inversely affect their appeal.

Visit frequency represents the volume of people visiting a location within a given period and serves as an indicator of a destination's popularity. However, while high visit frequency underscores a place's appeal, it can also introduce challenges such as overcrowding, long wait times, and a lack of personal space, all of which may diminish the overall visitor experience.

Highly crowded destinations often affect the decision-making process for potential visitors in significant ways. For travelers seeking relaxation or tranquility, the perception of overcrowding can be a deterrent, prompting them to choose less frequented locations. Additionally, the reduced quality of experience caused by overcrowding can manifest in logistical challenges, such as difficulties accessing key attractions or finding suitable accommodations. Environmental concerns linked to heavy footfall, including degradation of natural or urban spaces, further compound these issues, negatively influencing perceptions of the destination.

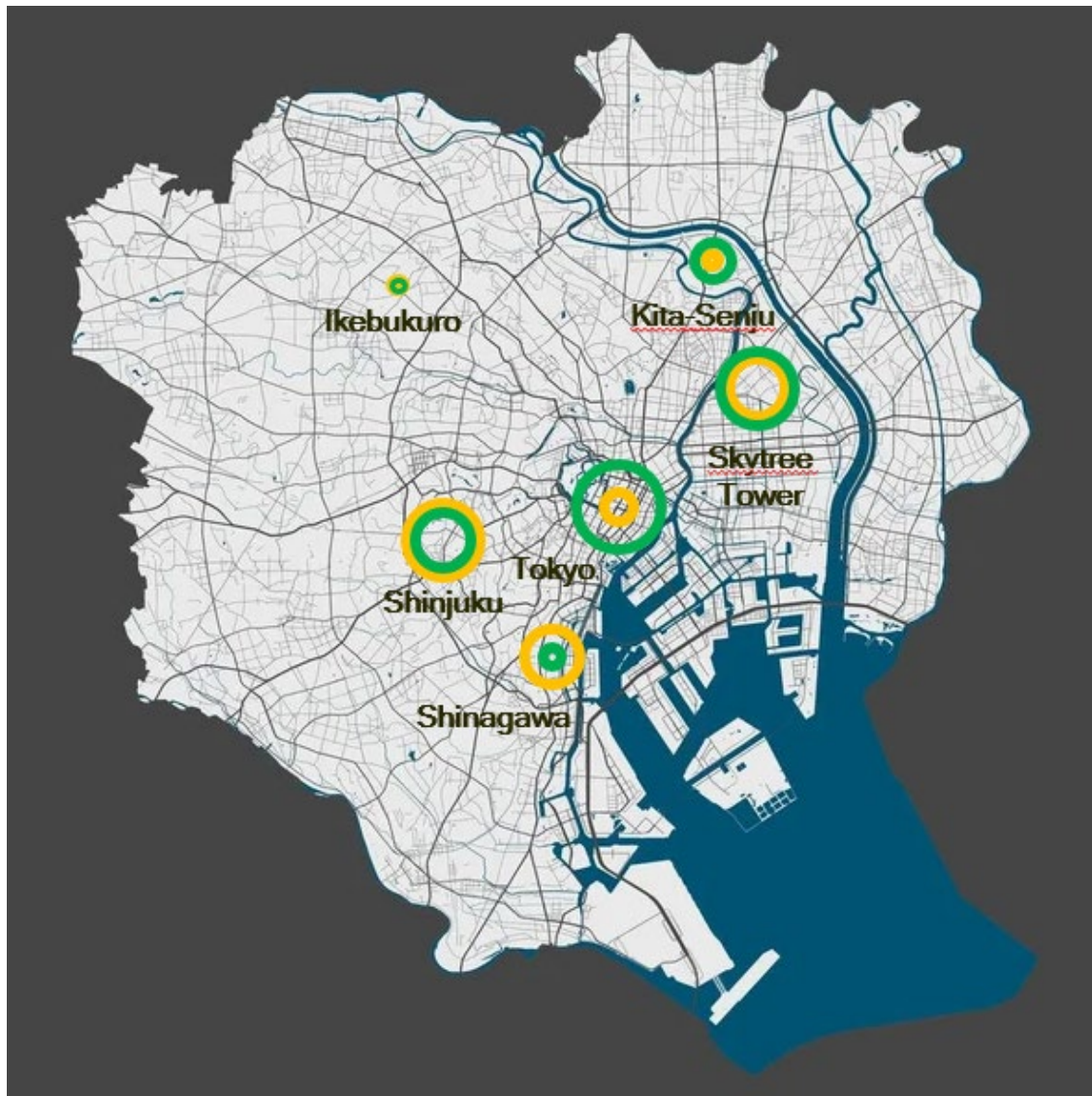


Fig. 2.1: Visualizing visit frequency and attractiveness of some Tokyo's spots: Kita-Senju, Tokyo Skytree, Shinagawa, Shinjuku, Tokyo, and Ikebukuro (Circle size proportional to values; Green for Attractiveness, Orange for Frequency)

Travelers often weigh the attractiveness of a destination against its visit frequency to assess the overall experience. A destination that is both appealing and not overcrowded is more likely to be selected. In contrast, destinations with excessive crowds may see a decline in visitor interest, as individuals opt for less crowded alternatives offering similar experiences.

2.2 Human visitation law

To better understand destination decision-making principle, researchers have deduced a universal human visitation law, which suggests that the likelihood of choosing a destination is a function of its attractiveness, balanced against the perceived challenges of high visit frequency and the distance from the current location.

The concept of the human visitation law describes how people distribute their visits across different locations, and how the frequency of those visits varies with distance and location attractiveness. This relationship is captured through a formula that defines the probability of selecting a particular location as a new destination.

The formula is:

$$\rho_i(r, f_i, \mu_i) = \frac{\mu_i}{(rf_i)^\eta} \quad (2.2)$$

Where :

- ρ_i represents the probability of selecting spot i as a new destination.
- r is the distance from the current location to spot i .
- f_i is the frequency of visits to spot i .
- μ_i is the attractiveness of spot i , which is a measure of how desirable or appealing the location is.
- η is a scaling exponent, typically around 2, which governs the probability of selecting a location decay with respect to the distance and visit frequency.

The formula indicates that the likelihood of choosing a particular spot i as a new destination is influenced by its attractiveness μ_i and is inversely related to the product of the distance r and the visit frequency f_i , both raised to the power of η . This means that:

- A more attractive location increases the chances of it being selected.
- The farther the location is, the less likely it is to be chosen.
- If a location has been visited more frequently, the probability of choosing it again decreases.

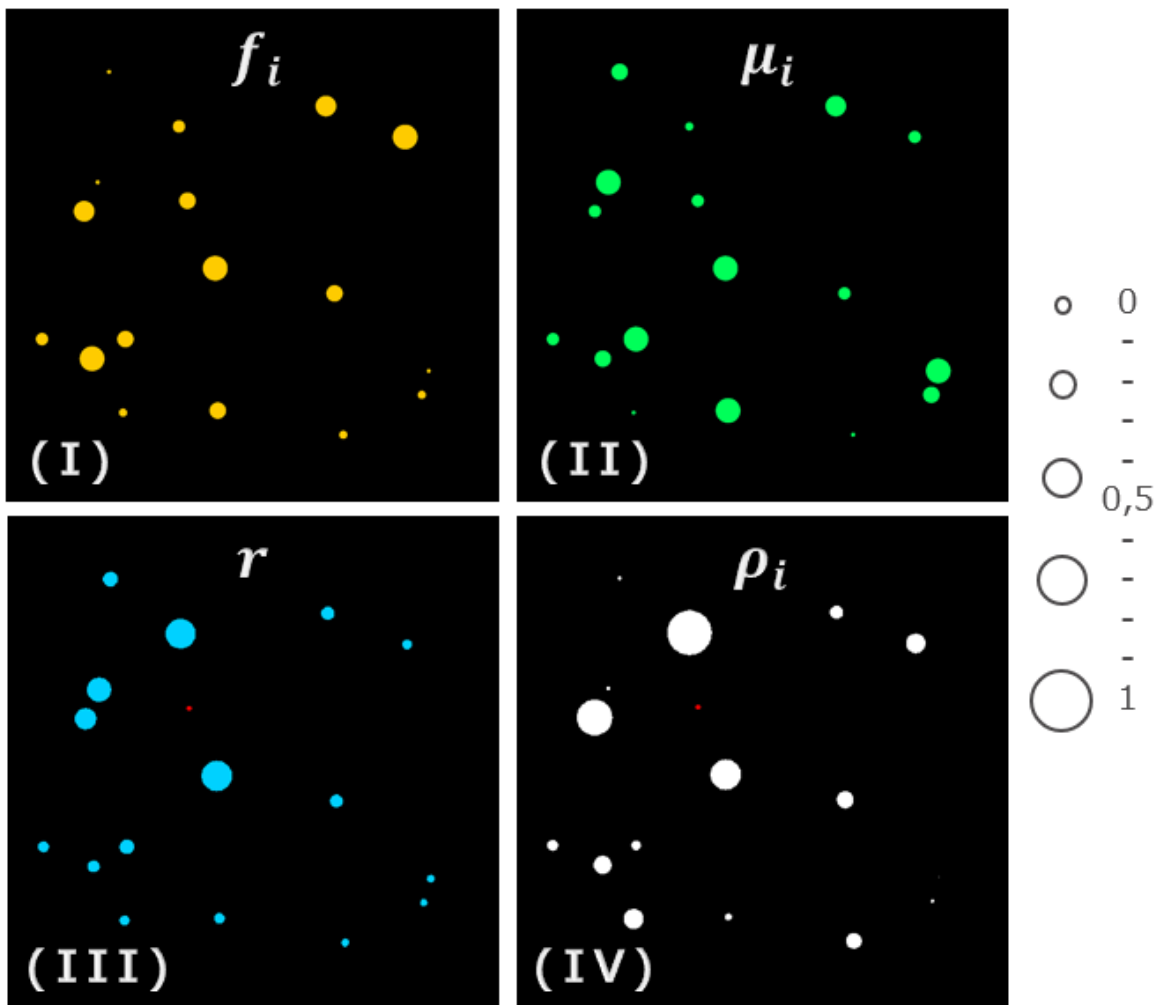


Fig. 2.2: (I) Random visit frequency f_i , (II) random attractiveness μ_i and (III) distance bias r influence on (IV) the probability ρ_i of a hotspot to be selected as new target relative to red spot location.

Figure 2.2 illustrates how different parameters—random visit frequency f_i , random attractiveness μ_i , and distance bias r —combine to influence the final probability ρ_i of selecting a hotspot as a new target relative to the current location. In simple terms, people are more likely to visit locations that are both attractive and nearby, while avoiding places that are farther away or have been visited too often. The scaling exponent η , typically around 2 [1], indicates that the likelihood of visiting a location diminishes rapidly as either the distance or the visit frequency increases. This principle provides insight into human mobility patterns and explains why certain spots in cities become hotspots, with people favoring nearby and less frequently visited locations.

2.3 Target selection algorithm

A bias-based target selection algorithm has been implemented and generates a random integer that adheres to certain constraints. Its purpose is to select an index from the set of hotspots, excluding the hotspot where agent is currently localized, while considering a weighted probability distribution. The algorithm accounts for the weights of each index, derived from the human visitation law, and uses a random float input to determine the selection in a probabilistic manner.

First, the algorithm calculates cumulative weights for all indices, excluding the one specified. This involves iterating through the list of hotspots and summing the weights based on conditions that ensure the exclusion of the specified index. Two cumulative weight variables track ranges—one up to the index before the excluded index, and the other including the excluded index. These cumulative weights define intervals for determining whether the randomly generated value falls within the range that would select the excluded index. If the random value lands in this range, the algorithm adjusts the selection to avoid the excluded index by returning the next valid index.

If the random value does not fall within the range tied to the excluded index, the algorithm iterates through all weights again, adding them cumulatively until the random value matches a particular interval. The index corresponding to this interval is returned as the selected target. This process ensures that all indices, except the excluded one, are selectable with probabilities proportional to their weights.

第3章 Multi-agent hybrid model

This chapter delves into the foundational aspects of the multi-agent hybrid model, exploring the key elements that define its structure, operation, and emergent behavior. The discussion is organized into three primary areas:

- Agent morphology
- Guiding principle
- From local information to self-organization

These sections collectively provide a comprehensive understanding of how individual agents interact and contribute to the emergence of system-level patterns.

3.1 Agent morphology

The morphology of an agent refers to its physical or conceptual structure, as well as its capabilities for perception, interaction, and action within the system. In this specific model, each agent is composed of three key components: a view area, a distance offset, and a view area width. These elements define how the agent perceives and interacts with its environment, guiding its behavior and decision-making.

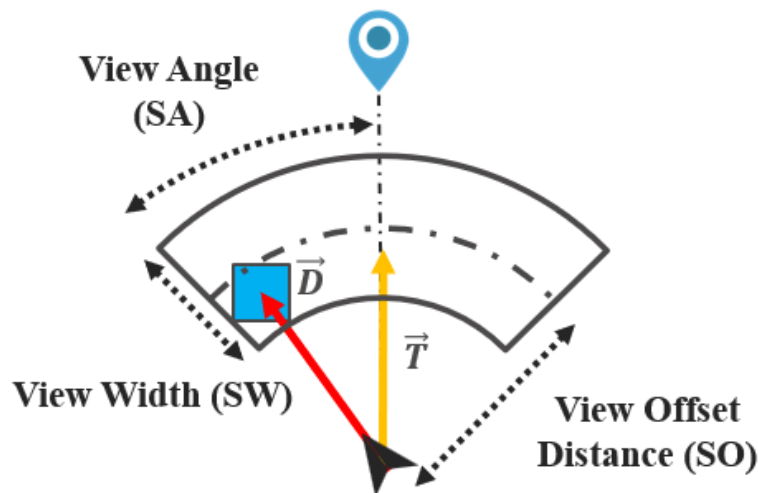


Fig. 3.1: Agent morphology with parameters.

The view area represents the spatial region within which the agent can observe and respond to its surroundings. This area determines the scope of environmental data the agent can process, influencing its ability to identify targets or respond to changes.

The distance offset specifies the relative position of the view area from the agent. This parameter allows the agent to focus on regions that are not immediately adjacent, enabling it to anticipate and react to environmental features or interactions further afield.

The view area width defines the angular breadth of the agent's field of view. A narrower view limits the agent's focus to a specific direction, promoting precise and targeted behavior. Conversely, a wider view enables the agent to take a broader range of environmental factors into account.

The inclusion of this focus ability parameter enriches the model by adding a layer of flexibility and adaptability to agent behavior. It allows for the exploration of scenarios where agents must dynamically balance focus and awareness, depending on their objectives and environmental demands. This adaptability leads to more realistic simulations, as real-world entities often need to shift between narrow focus (e.g., pursuing a target) and broader awareness (e.g., scanning for threats or opportunities).

In a self-organizing system, the ability to modulate the view area width can enhance emergent behaviors. For instance, agents with a narrow focus might excel at goal-directed tasks, while those with a wider focus might contribute to group-level coordination and environmental exploration. This diversity in agent behavior, enabled by the focus ability, drives richer and more robust emergent patterns in the overall system.

Together, these morphological elements equip the agent to operate autonomously and dynamically within the system. By perceiving its immediate environment through its view area, adjusting its focus with the distance offset, and modulating its attention using the view area width, the agent forms the basis for complex, coordinated behaviors that emerge at the system level.

3.2 Guiding principle

The movement of agents is governed by a principle that balances environmental stimuli and target-oriented navigation. Agents detect and respond to pheromones deposited in their view area, guiding their trajectory based on the calculated relevance of the detected stimuli. This mechanism ensures that the agents adapt dynamically to their environment while maintaining focus on their ultimate destination.

When an agent detects pixels containing pheromones within its field of vision, it evaluates the weight of each pixel using the following equation:

$$w_i = p_{aff}(i) \frac{1 + \vec{T} \cdot \vec{D}}{2}$$

Where:

- w_i : The resulting weight of pixel i .
- $p_{aff}(i)$: The affordance value of pixel i , representing the pheromone concentration.
- $\vec{T}$: The agent's target direction vector.
- $\vec{D}$: The direction vector of pixel i relative to the agent's position.

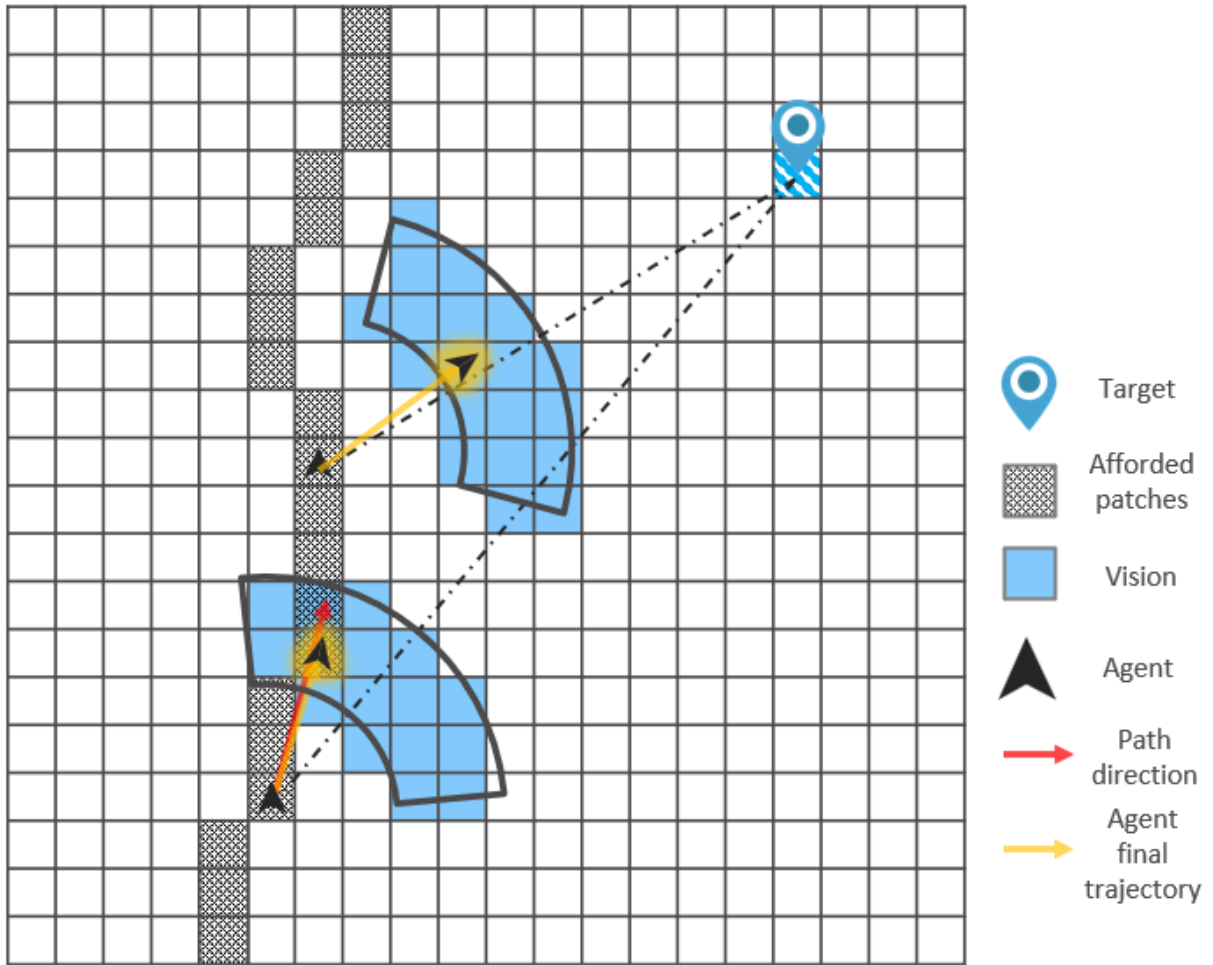


Fig. 3.2: Guiding principle of agent movement to create desire path phenomenon.

This formulation combines the pheromone concentration with the alignment of a pixel' s direction toward the agent' s target, favoring paths that lead closer to the intended goal.

Once the weights for all pixels are calculated, the agent determines the average direction from the weighted pixels, which guides its next movement. In the absence of detectable pheromones, the agent proceeds straight along its current target direction. This dual focus on local stimuli and long-term navigation ensures that agents adapt dynamically to their surroundings while maintaining a steady progression toward their objectives.

A critical component of this system is the dissipation of pheromones over time. Pheromone concentrations gradually decrease, reflecting the loss of relevancy of unused paths. This mechanism ensures that the path system is not static, allowing it to evolve and adapt as environmental conditions change. By enabling the natural decay of outdated trails, the system prevents agents from being drawn to obsolete routes and promotes the exploration of more effective paths. This dynamic interaction between pheromone deposition and dissipation fosters a balance between reinforcing well-used pathways and encouraging adaptive responses to shifting stimuli, ultimately enhancing the system's ability to optimize navigation in complex environments.

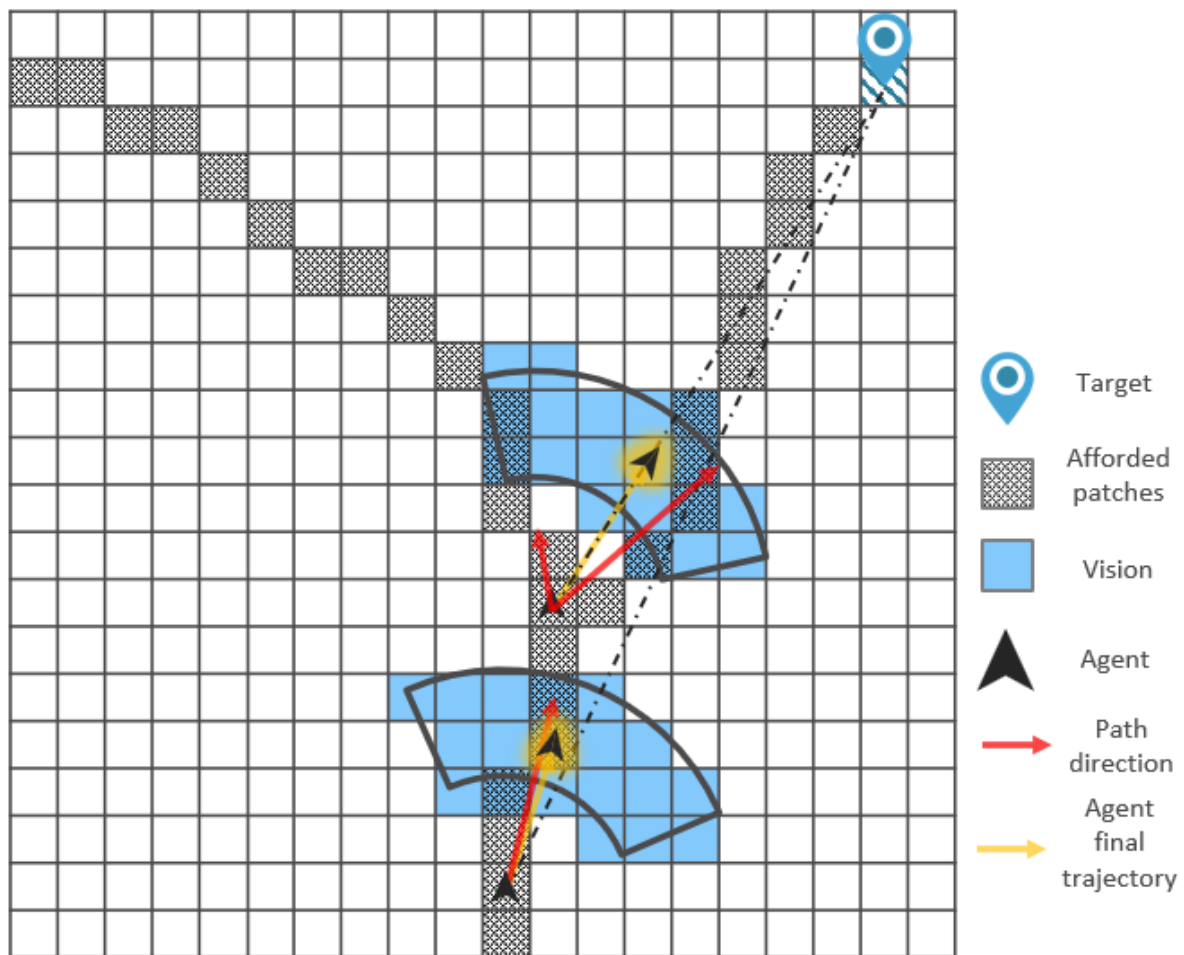


Fig. 3.3: Illustration of the agent behavior adaptation to multiple paths.

3.3 From local information to self-organization

The behavior of individual agents, governed by simple local interactions, can collectively lead to the emergence of global self-organization within a system. Each agent operates autonomously, sensing its immediate surroundings, responding to environmental stimuli such as pheromones, and making decisions based on localized information. These decentralized interactions are influenced by the deployment of a large number of agents, which amplifies the effect of individual actions on the overall system.

As agents deposit pheromones along their paths, they create trails that encode their movements and decisions. Other agents sense and respond to these pheromone trails, reinforcing well-traveled paths while exploring less-visited regions. This positive feedback loop strengthens effective routes, allowing the system to converge on paths that provide optimal connectivity between important locations. At the same time, pheromone dissipation prevents the system from becoming overly rigid, enabling it to adapt dynamically to changing conditions or obstacles.

This self-organizing behavior is particularly powerful for creating optimized path systems. By deploying a large number of agents, the system can efficiently explore vast and complex environments, identifying multiple potential routes. Over time, the interaction between agents and their environment naturally selects the shortest, most efficient, or most resource-effective paths. The emergent paths are not pre-defined but arise from the collective actions of agents, making the system highly robust and adaptable.

第4章 Simulation and results

The simulation environment shown in Fig. 4.1 and 4.2 was established to perform the multi-agent model ability on a 2D texture map. In this part, we run the model over 500 generations and we evaluate the performances of the following items are discussed in this chapter:

- Example of different paths system solutions
- Paths relevancy and traffic density
- Convergence
- Paths system performances
- Evaluation of different agent morphology configurations

4.1 Examples of resulting paths system solutions

The model demonstrates its ability to generate various types of network structures, including direct spanning trees, Steiner trees, and intermediate tree configurations.

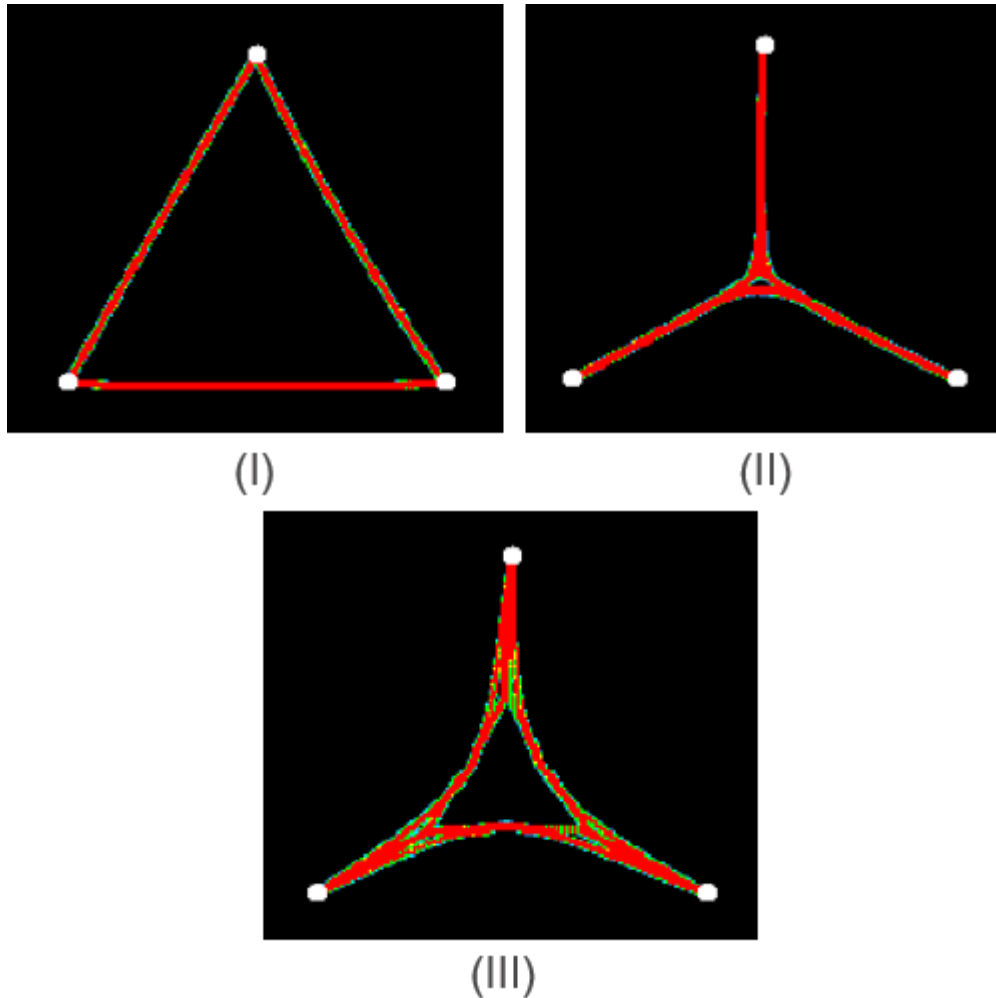


Fig. 4.1.1: Example of resulting solutions for three points arranged in a triangular configuration, with parameters pair {view angle, view distance}:
(I) Direct spanning tree solution {20, 8}, (II) Steiner solution {300, 30}, (III) Intermediate solution {140, 30}

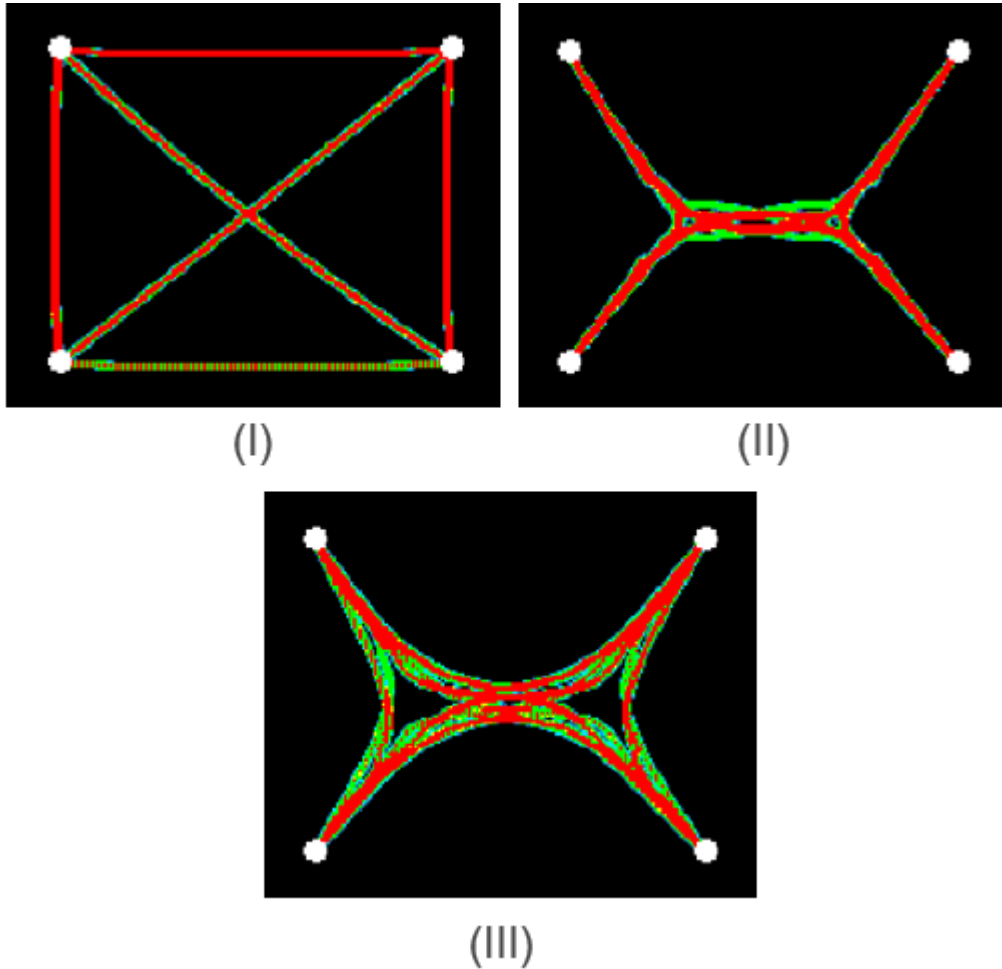


Fig. 4.1.2: Example of resulting solutions for three points arranged in a square configuration, with parameters pair {view angle, view distance}:
(I) Direct spanning tree solution {20, 8}, (II) Steiner solution {300, 30}, (III) Intermediate solution {140, 30}

As illustrated in Fig. 4.1 and Fig. 4.2, the model can adapt to different spatial configurations, such as triangular and square arrangements, to produce efficient network patterns. In a triangular configuration, the model is able to generate connections that minimize the total distance between nodes, effectively forming direct spanning trees when direct routes are prioritized. Similarly, Steiner trees can emerge when intermediate nodes are introduced to further optimize the network by reducing the overall path length, balancing efficiency and simplicity.

The resulting patterns also depend significantly on the parameters governing the agents' view angle and view distance offset. These parameters influence the way agents sense and interact with their environment, shaping the network's structure. This, in turn, impacts the performance of the paths system in terms of the number of complete travels achieved within a limited time period and the total length of the network. By fine-tuning these parameters, it is possible to balance the efficiency of travel with the overall resource investment in the network.

In a square configuration, the model demonstrates flexibility in forming both direct spanning trees and intermediate tree patterns, depending on the spatial relationships, stimuli distribution, and parameter settings. The intermediate tree solutions provide a balance between directness and resource optimization, enabling the system to adapt to changing demands or constraints.

These results highlight the versatility of the model in addressing different network optimization problems, making it a valuable tool for applications ranging from transportation networks to communication systems. The ability to transition seamlessly between direct, Steiner, and intermediate tree solutions underscores its robustness, while the influence of parameter settings offers opportunities for tailoring the network to specific performance goals.

4.2 Paths relevancy and traffic density

In the resulting path system, the accumulation of pheromones serves as a key indicator of path relevancy and traffic density. Paths with higher pheromone concentrations reflect frequent use, signifying their importance within the system. This mechanism allows for the identification of heavily trafficked routes and the prioritization of these paths in the network's structure and optimization.

The dissipation of pheromones over time plays a crucial role in reevaluating the usage of paths. This natural decay simulates the loss of relevancy for unused or underutilized paths, enabling the system to adapt dynamically to evolving conditions. As paths lose their pheromone markings, agents explore alternative routes, ensuring that the final path system reflects current and efficient patterns of movement.

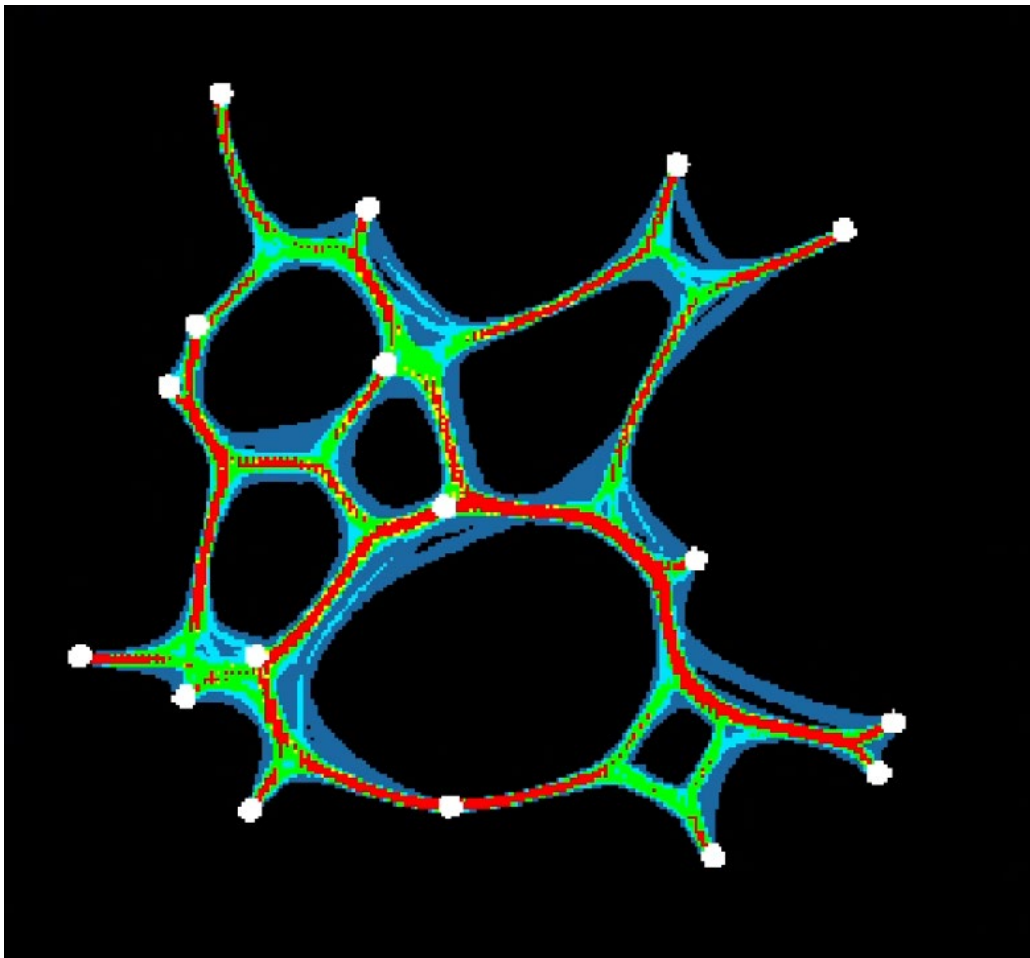


Fig. 4.2: Traffic density map of the resulting network after 500 generations.

The variation in traffic density across the network provides valuable insights for infrastructure planning and resource allocation. High-density paths, marked by significant pheromone accumulation, suggest the need for robust and high-capacity infrastructure, such as railways or multi-lane highways, to efficiently handle the flow of agents or vehicles. Conversely, lower-density paths indicate less critical routes that may require minimal investment, such as bicycle lanes, pedestrian walkways, or single-lane roads for light traffic.

This adaptive evaluation of path relevancy and traffic density offers a scalable approach to designing optimized transportation systems. It allows decision-makers to allocate resources proportionally, ensuring that infrastructure development aligns with actual usage patterns. Furthermore, this model can be applied to urban planning, logistics, and autonomous systems, creating dynamic and efficient networks that evolve with the needs of their users.

4.3 Convergence of the model

During the simulation, the path system continually readapts and evolves over time, gradually moving toward a state of convergence. In the example of deploying 1,000 agents with a view angle of 160 degrees and a view distance of 10 pixels within a 300x300 pixel 2D texture map containing 17 random hotspots, we observe a distinct trend: while the outer contours of the path system stabilize, its internal structure continues to exhibit dynamic changes.

In our example, it is evident that the bottom-right section of the path system is the first to converge to a stable solution. This area contains hotspots that are relatively isolated from the rest of the network, making it simpler for the system to identify and establish an optimal configuration. The isolation minimizes interactions with other parts of the network, allowing a stable solution to emerge more quickly.

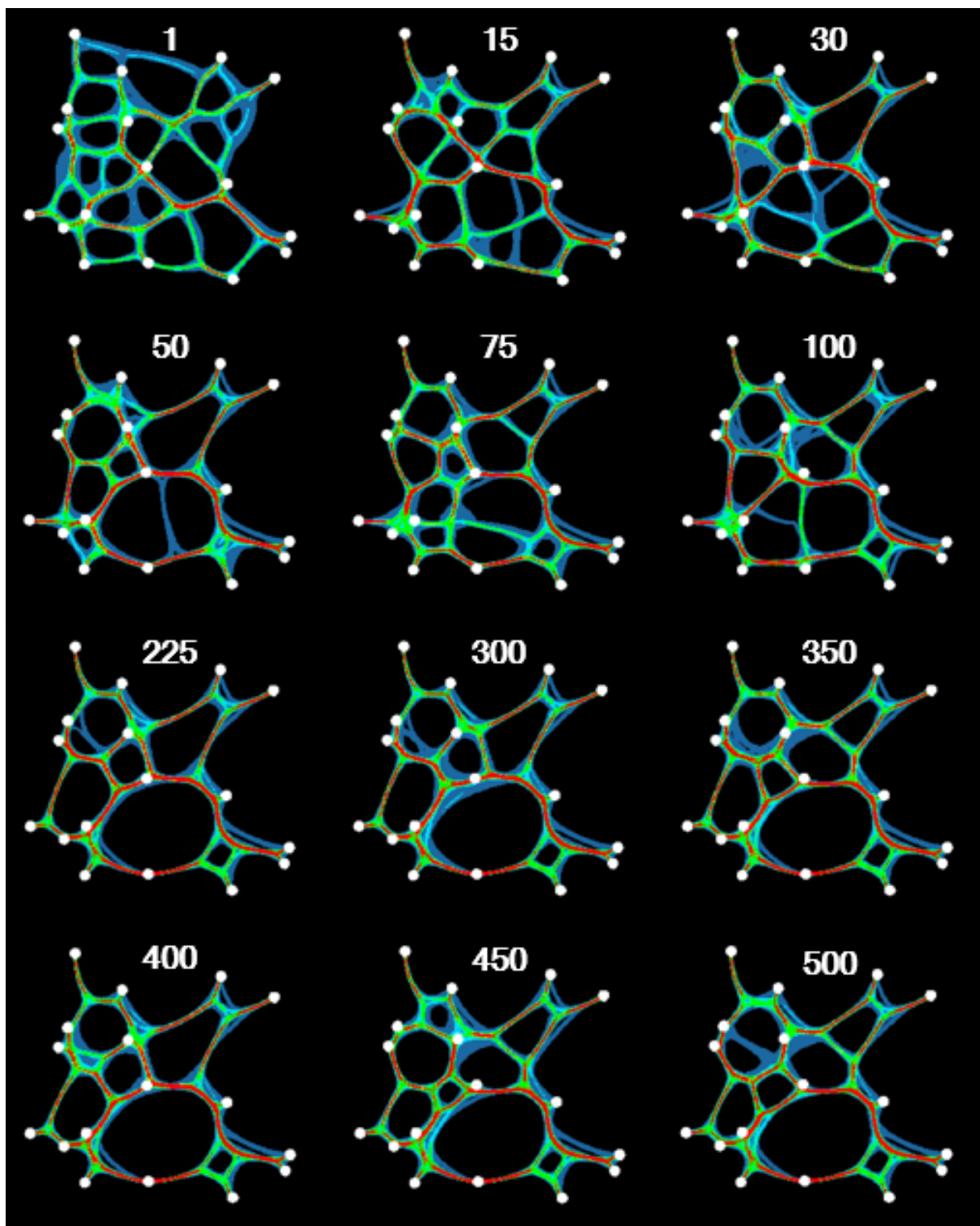


Fig. 4.3.1: Evolution of paths system over 500 generations. The right part of the pattern highlights the relevancy of trails through continuity of traffic density along path and low-traffic-density roads adjacency to heavily traveled main road reproduce phenomenon of small desire paths emergence along urban designed roads.

Once this stable region is established, the rest of the pattern begins to adapt and reorganize. The stability of the bottom-right section serves as an anchor, influencing the ongoing development of the rest of the paths system. This sequential adaptation highlights the model's ability to progressively refine its structure, starting with simpler, more isolated regions and then extending to more complex and interconnected areas.

From Figure 4.3.1, we can observe that the red continuous trail on the bottom-right remains intact through generations, underscoring its relevance in the final pattern of the path system. This persistent high-traffic-density trail demonstrates its importance as a stable and integral part of the network. Conversely, the inner regions of the path system remain in a state of continuous readaptation. These inner areas are more sensitive to the inherent randomness of agent behavior, such as agents selecting random destinations and the timing of their use of specific trails. This heightened sensitivity leads to ongoing fluctuations and restructuring within the interior of the path system.



Fig. 4.3.2: Example of desire path adjacent to traveled main road in simulation (left) and in nature (right).

The simulation highlights another significant phenomenon: the emergence of low-traffic-density roads adjacent to heavily traveled main roads with high traffic density. This mirrors what is often observed in nature, where small desire paths appear alongside urban-designed roads, as shown in Figure 4.3.2. These secondary trails serve as adaptive, lower-cost alternatives to the primary paths, reflecting the system's ability to accommodate varying traffic needs and agent behaviors.

Traffic density map provides additional insights into this process. High-traffic-density trails, represented in red, are the most stable and resilient paths, persisting through multiple iterations. Conversely, trails with low traffic density are more susceptible to changes, often merging with other trails, shifting shape, or disappearing entirely over time. The continuity of traffic density along a trail further underscores its long-term relevancy, as stable and well-used trails are less prone to modification.

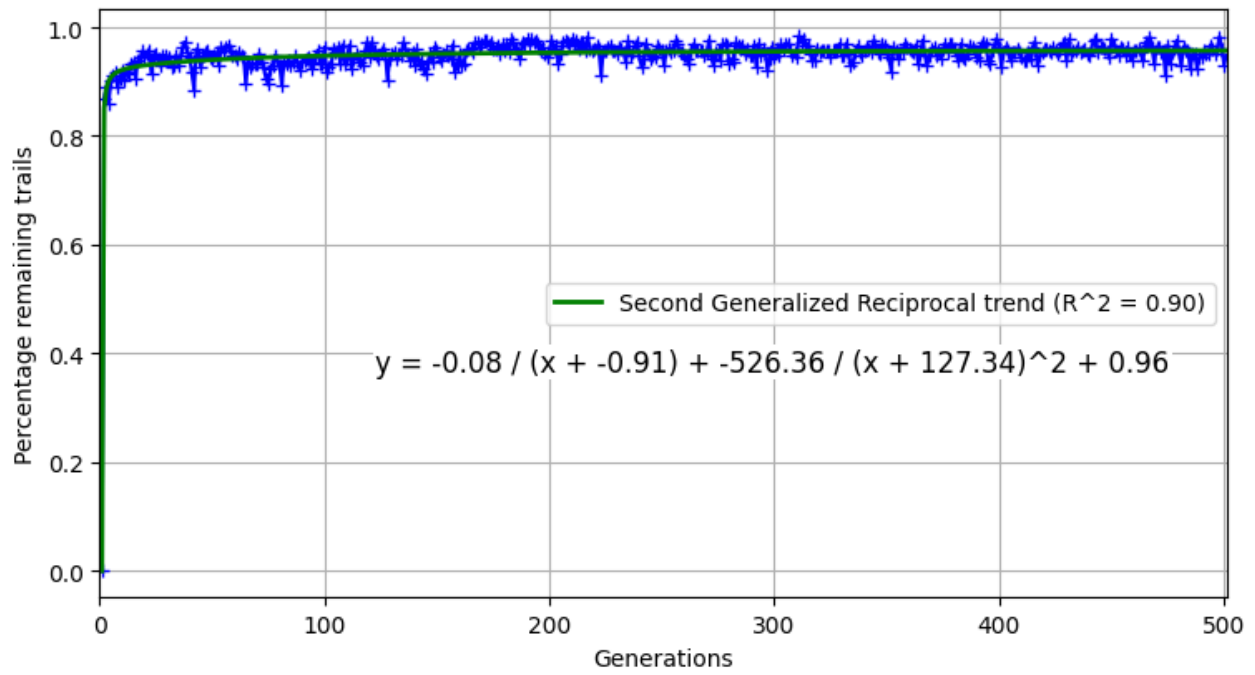


Fig. 4.3.3: Percentage of remaining trails through generations.

Figure 4.3.3 demonstrates the evolution rate of this simulation, which is approximately 4%, indicating that the overall shape of the path system changes by only 4% between each generation, which depends on the time length of one generation in terms of number of iterations. This low evolution rate highlights the adaptation ability of the model as well as the potential deadlocks in finding a stable pattern solution.

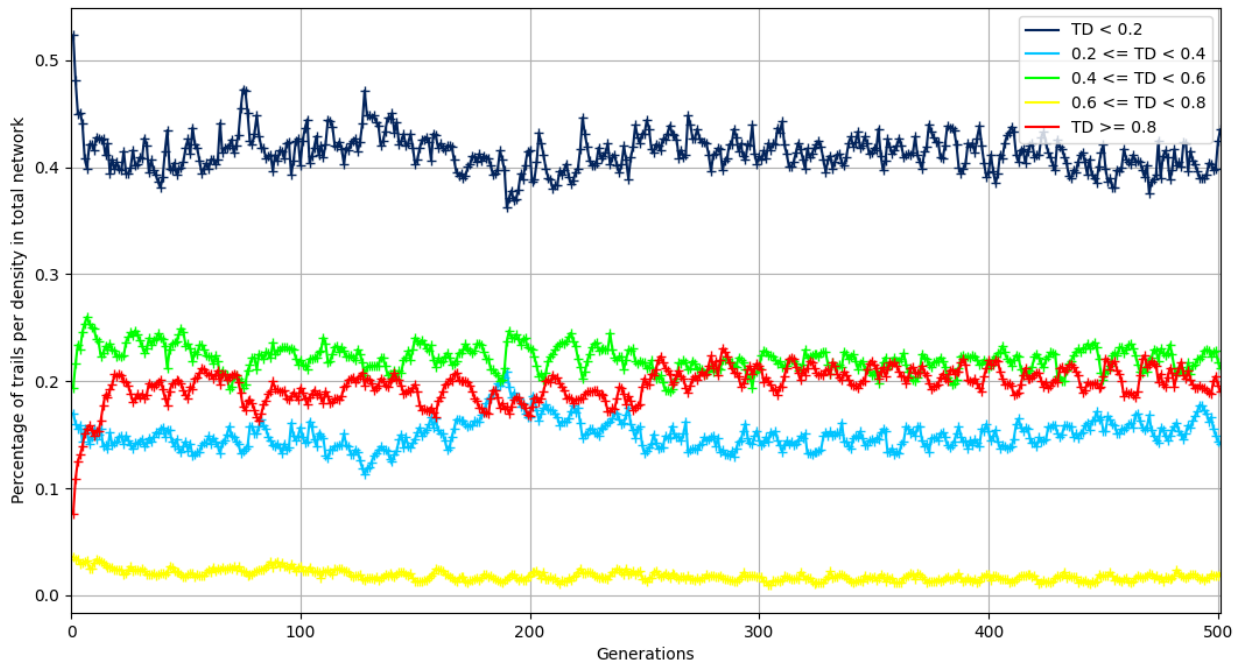


Fig. 4.3.4: Percentage of traffic density distribution through generations.

The proportion of traffic density levels within the path system is highly influenced by the initial parameters of the simulation, such as agent behavior and environmental setup. Nevertheless, the model consistently demonstrates its ability to achieve a relative equilibrium in traffic density distribution while retaining the capacity for self-reshaping and optimization. This dynamic balance ensures that the path system remains efficient, adaptive, and capable of reflecting real-world patterns of road and trail usage.

4.4 Paths system performances

In this section, we outline the key parameters used to evaluate the performance of a path system. The discussion is structured around three core aspects:

- Paths efficiency
- Complete travels
- Total size of system

4.4.1 Path efficiency

The efficiency of road networks is a critical aspect of evaluating transportation systems. Efficiency can be assessed for individual routes and the overall path system by examining the relationship between the direct route distance and the travel distance along constructed roads. While a single route may appear less efficient compared to its direct counterpart, the shared use of roads within a larger network often minimizes the total length of the road system, providing a more resource-efficient solution overall.

Each route in the system has a specific efficiency value, e_i , which is calculated as:

$$e_i = \frac{\text{Direct route Distance}_i}{\text{Travel Distance}_i}$$

This metric reflects how closely a route aligns with the shortest possible path between two points. A perfectly direct route would have an efficiency of 1, meaning the travel distance equals the direct route distance. Lower values indicate greater deviations from the direct path, often resulting from the constraints of shared road usage, terrain, or urban planning considerations.

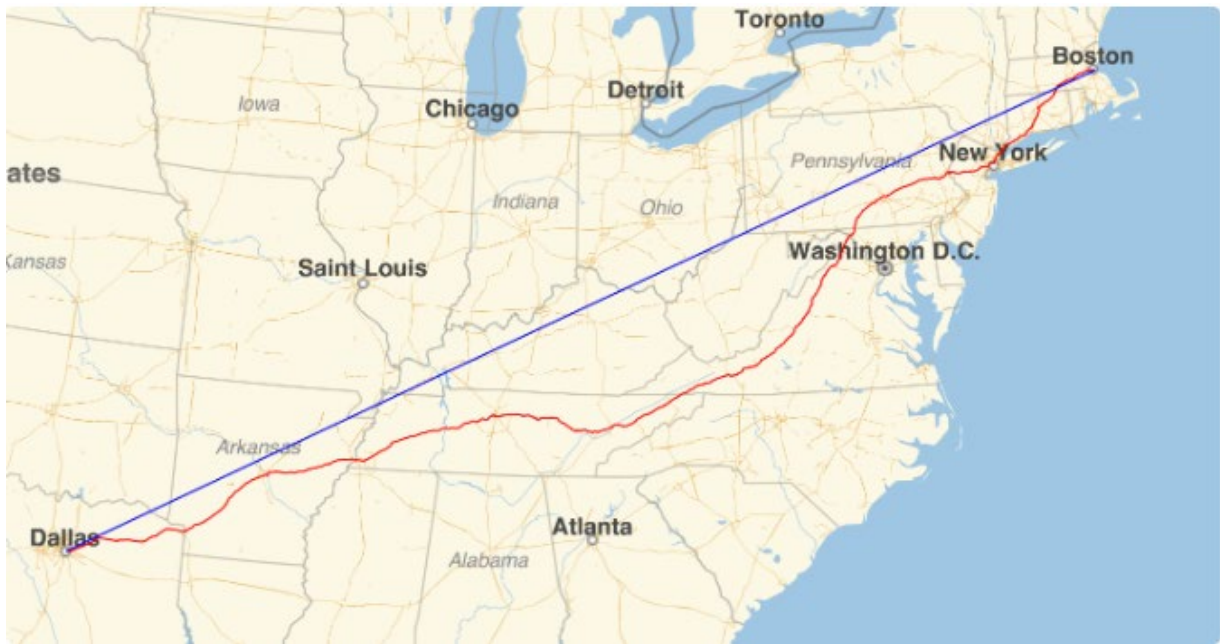


Fig. 4.4.1: Travel route (red) and direct route (blue) between Dallas and Boston.

In real-world applications, road networks rarely consist of perfectly direct routes. Instead, paths are optimized for shared usage and resource allocation, balancing efficiency with practicality. For example, constructing perfectly direct roads between every pair of destinations would lead to an unsustainable increase in total road length and construction costs. By sharing segments of roads across multiple routes, the overall length of the road system is reduced, even if individual routes deviate from the most direct paths.

In this research, this balance between individual route efficiency and system-wide optimization is a fundamental focus. The modeling framework I developed demonstrates how the dynamic interactions of agents and pheromone-based mechanisms can create self-organized networks that prioritize shared usage while maintaining acceptable levels of individual route efficiency. The resulting networks strike a balance between minimizing the total system length and ensuring that travel times remain competitive.

This approach provides insights into how transportation systems can be designed to maximize resource efficiency while maintaining high functionality. By understanding and quantifying route efficiency and its relationship to the larger system, we can better inform the development of road networks and other shared infrastructure systems.

4.4.2 Complete travels

A critical parameter for evaluating the performance of a path system is the total number of complete travels achieved within a limited time period. This parameter takes into account:

1. The number of agents deployed (N).
2. Each agent's speed (v).
3. The set of routes traveled by each agent during the evaluation period.
4. The efficiency (e_i) and distance (d_i) of each route.
5. The time length of the evaluation period (t).

For a single agent that travels across multiple routes, the total number of complete travels is determined by summing the contributions of each traveled route. If the agent travels a set of m routes ($i = 1, 2, \dots, m$), the formula becomes:

$$T_{agent} = \sum_{i=1}^m \frac{v \cdot t \cdot e_i}{d_i}$$

Where:

- e_i is the efficiency of the i -th route.
- d_i is the travel distance of the i -th route.
- v is the agent's speed.
- t is the evaluation time period.

This formula accounts for the fact that the agent may complete partial travels on longer routes, depending on the time available.

For the entire system with N agents, the total number of complete travels is the sum of the contributions from all agents. Since each agent may travel different sets of routes during the time period, the system-wide formula becomes:

$$T_{system} = \sum_{j=1}^N \sum_{i=1}^{m_j} \frac{v \cdot t \cdot e_{i,j}}{d_{i,j}}$$

Where:

- m_j is the number of routes traveled by the j -th agent.
- $e_{i,j}$ and $d_{i,j}$ are the efficiency and distance of the i -th route traveled by the j -th agent.

The formula provides valuable insights into the behavior of agents and the design of an effective path system. The diversity of routes traveled by agents reflects their adaptability and highlights the flexibility of the network. Routes with higher efficiency values enable agents to complete more travels, thereby enhancing overall system performance. Additionally, shorter travel distances allow agents to achieve a greater number of trips within the same time period, emphasizing the importance of minimizing distances in path system design.

Some paths can be used multiple times and by multiple agents, which is reflected in the traffic density and paths relevancy map.

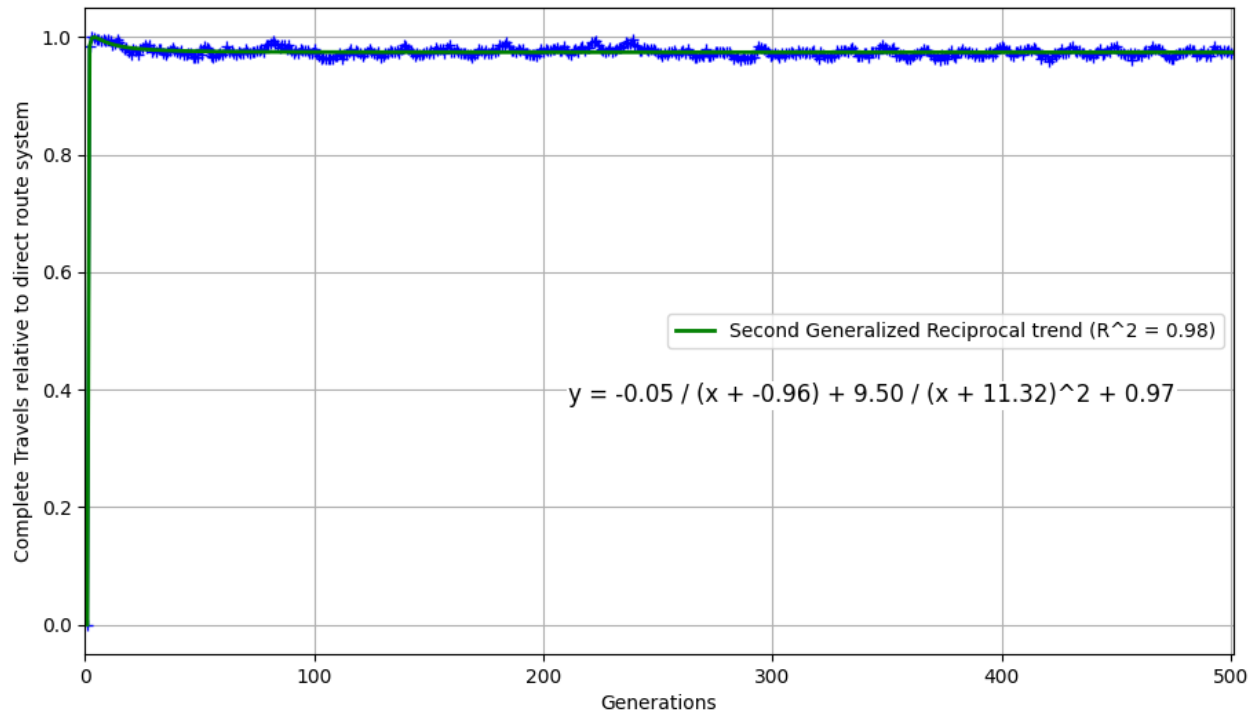


Fig. 4.4.2: Evolution of complete travels of the system through generation. One generation corresponds to 6000 iterations in the simulation.

In our example, we observe a convergence in the number of complete travels as the path system stabilizes over time. The final number of complete travels converges to approximately 97% of the result obtained from the path system in the first generation.

4.4.3 Total size

The total size of the path system serves as another critical parameter for evaluating its performance. Over the course of the simulation, as the system adapts and evolves, it undergoes a process of shrinkage, reducing its overall size. This reduction in total length is particularly advantageous in contexts such as urban planning, where minimizing the extent of transportation infrastructure is crucial for reducing maintenance costs and optimizing resource allocation.

This shrinkage reflects the model's capacity to identify and retain only the most relevant and efficient routes while eliminating redundant or underutilized paths. By doing so, the system not only enhances its efficiency in terms of traffic flow and agent travel but also reduces the financial and environmental burden associated with maintaining an extensive transport network.

Figure 4.4.3 illustrates the evolution of the total size of the path system in our example. The data reveals that the constant readaptation inherent in the model makes it challenging for the total size to achieve complete convergence.

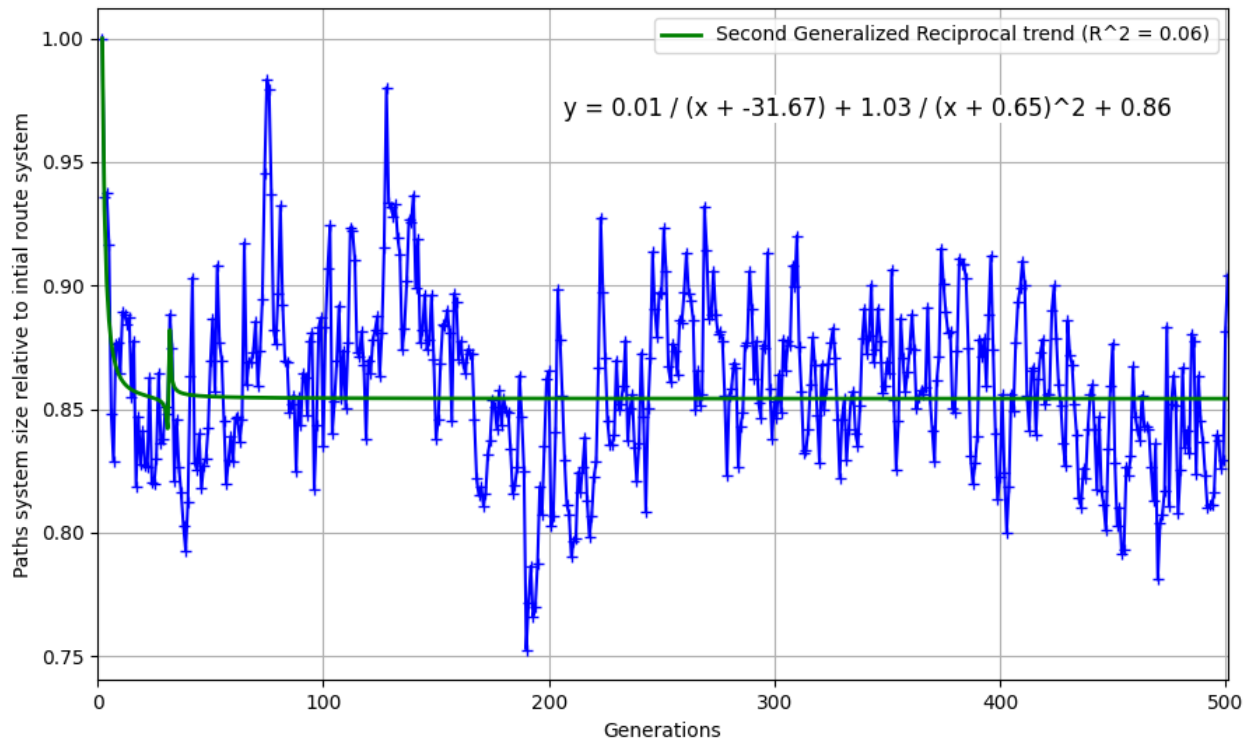


Fig. 4.4.3: Evolution of total size of path system through generations.

In our example, the system exhibits a variance of 22%, reflecting the dynamic adjustments of the network as it adapts to agent behavior and environmental conditions.

Despite this variability, the total size appears to stabilize around an average value, approximately 86% of the size of the initial configuration. This suggests that while the system continuously reshapes itself in response to changing conditions, it tends to gravitate toward a relatively consistent scale.

This behavior highlights the model's ability to balance adaptation and stability, maintaining an optimized network size while allowing for dynamic adjustments. The fluctuations in total size also underscore the model's sensitivity to parameters and agent interactions, which influence the system's capacity to streamline paths while preserving functionality and efficiency.

4.4.4 Texture resolution impact on simulation

The resolution of the texture in the simulation has a significant impact on the results and their analysis, as it directly influences the granularity of the paths and the precision of the generated patterns. To investigate this effect, we calculated the total size of the resulting path systems for two configurations: the Steiner pattern and the direct route pattern. The comparison was conducted by plotting the ratio of the total sizes of these two patterns across various texture resolutions and comparing it to the theoretical ratio derived from geometric models.

The plotted ratios reveal a trend where the simulated results approach the theoretical ratio.

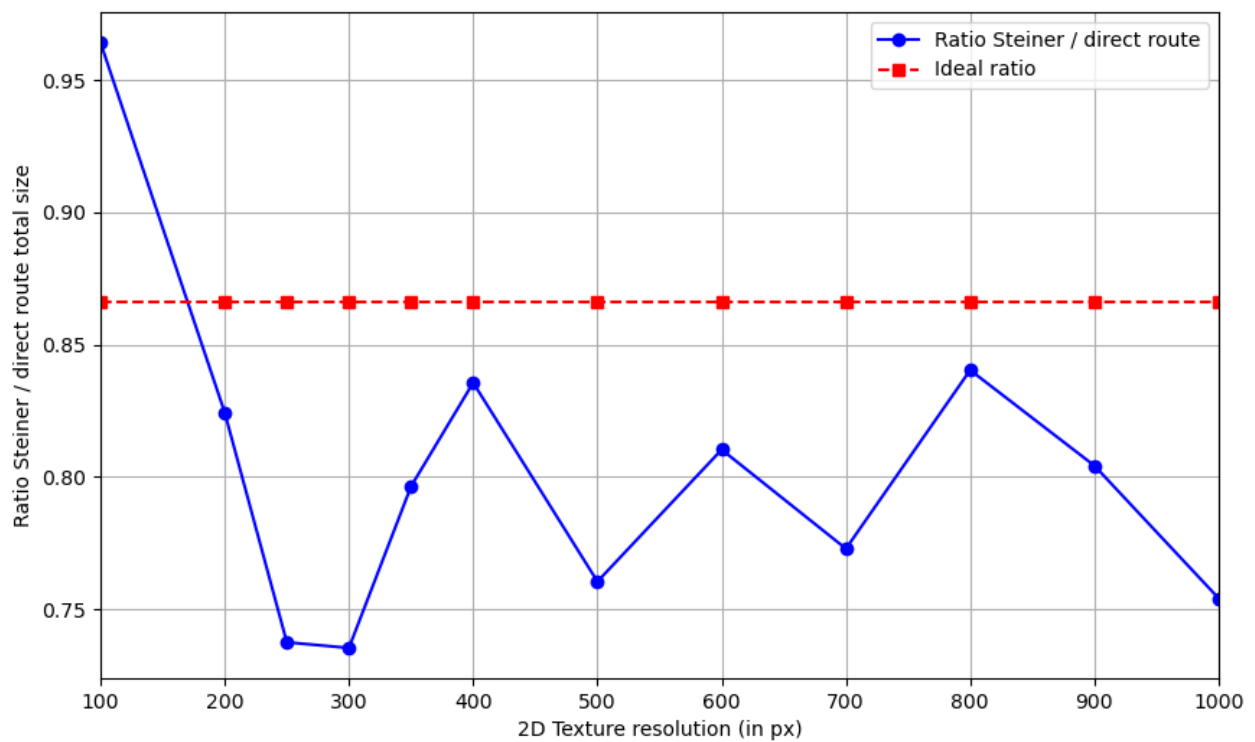


Fig. 4.4.4: Proportion between Steiner tree total size and Direct route tree size according to resolution of 2D texture in pixels. Hotspots are set in the equilateral triangular configuration.

第5章 Agent configuration evaluation

This chapter covers the different performances of different agent configurations, especially according to depth view and view angle. The discussion is organized into those three elements:

- Complete travels
- Path system size
- Evaluation limitations

5.1 Path system size

I ran a series of simulations with different configurations of agent morphology, specifically varying the view angle and depth view. The simulations were conducted over 30 generations of 5000 iterations each, on a 400x400 2D texture, utilizing an equilateral triangular hotspot configuration. The primary objective was to track the number of complete travels achieved within the limited time frame.

From the resulting data, we can identify relative distinct zones in the graph that correspond to various path types, such as Steiner trees, direct routes, and intermediate stable solutions, as illustrated in Figure 5.1.

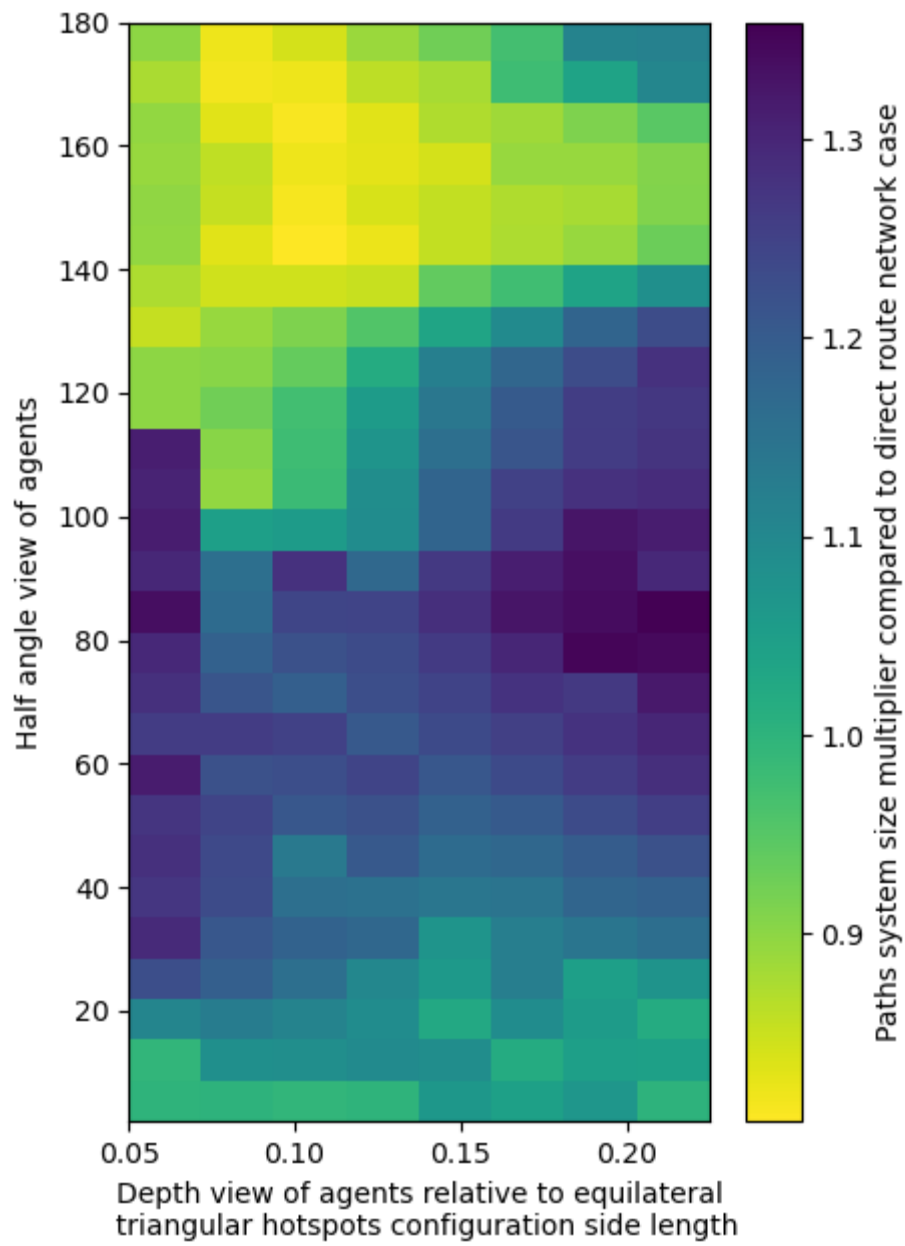


Fig. 5.1: Evolution of path system size performance relative to the direct route path system performance. The evolution is according to different combinations of angle and depth view.

The clearest areas in the graph correspond to configurations where the resulting paths forming Steiner trees. These areas indicate that agents, with high depth and a large view angle, were able to analyze their surroundings effectively and create overlapping paths, optimizing the overall system by connecting multiple hotspots efficiently. This suggests that the agents were capable of sharing roads and collaborating in the creation of a more interconnected and optimized structure.

Conversely, the darker areas in the graph represent configurations where more direct route patterns were formed. These paths were created when the agents had a lower view angle and depth, focusing on the most direct path to their target. In these configurations, agents prioritized the shortest routes, resulting in simpler path patterns with less optimization.

In addition, intermediate stable solutions began to emerge from the interaction of agents with varying view capabilities. These solutions reflect a balance between direct routes and Steiner trees, where paths evolve over time and optimize.

5.2 Complete travels

The performance of complete travels varies significantly across different path configurations, highlighting the trade-offs inherent in the system. When comparing Steiner tree patterns to direct route configurations, we observe notable differences in the number of complete travels achieved over the simulation period.

In a Steiner tree configuration, individual paths tend to be longer compared to direct routes. This is because Steiner trees prioritize the optimization of the overall system by creating shared paths that connect multiple hotspots. While this approach reduces the total length of the path system and minimizes redundancy, it inherently increases the travel distance for individual agents. Given that the velocity of agents and the length of each generation time remain constant throughout the simulation, longer paths result in agents taking more time to complete a single travel. Consequently, the total number of complete travels achieved in the same time period is reduced in Steiner tree configurations.

In contrast, direct route configurations focus on the shortest path between each agent's origin and destination. These simpler, more direct paths allow agents to complete their travels more quickly, leading to a higher total number of complete travels within the same simulation period. However, this increase in individual efficiency comes at the expense of overall system optimization, as the path system becomes less compact and more redundant.

From Figure 5.2, we can observe that the series of simulation give the same performance independently from the depth view. Moreover, the change of depth view allows the model to give a free-scale ability, without impacting the number of complete travels.

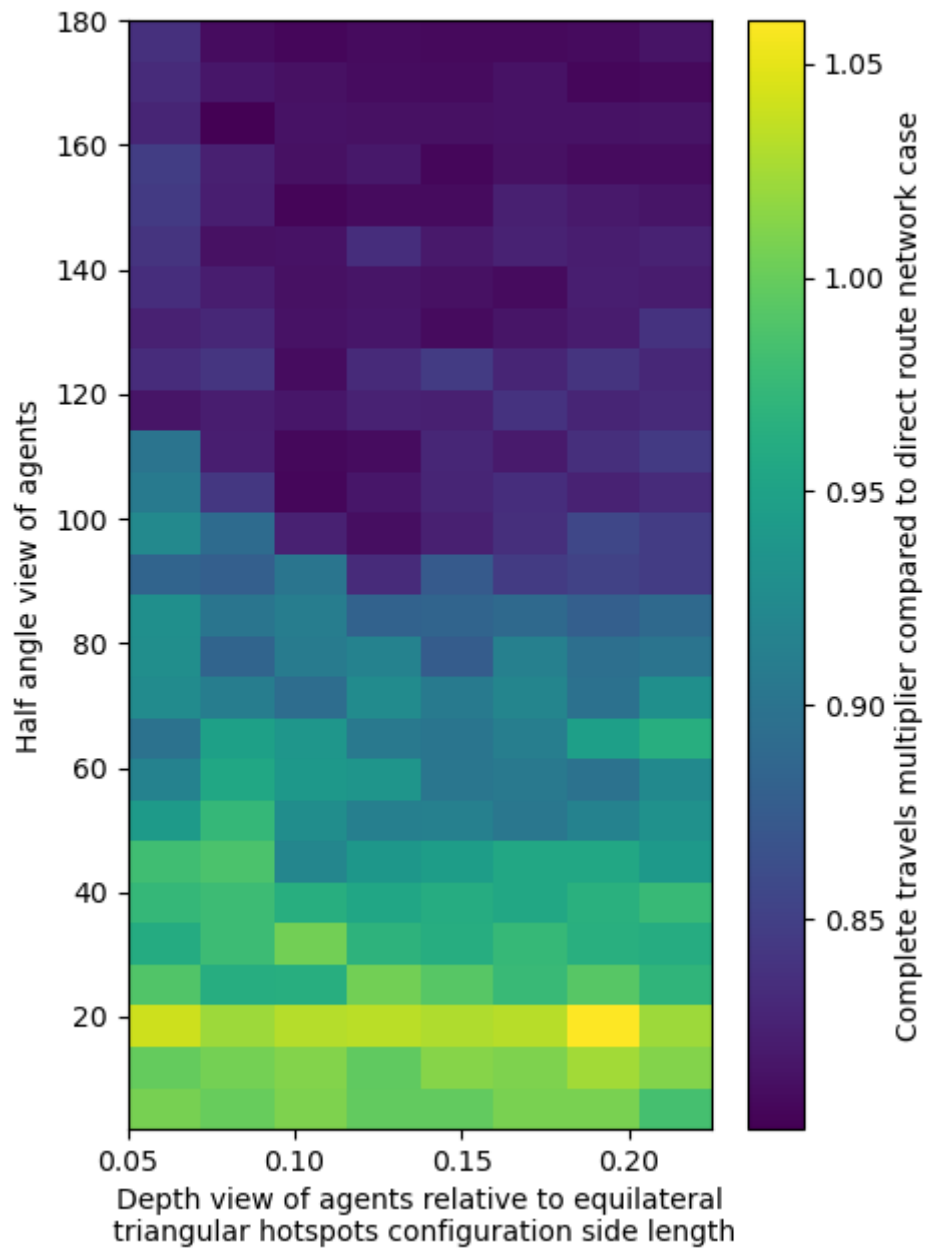


Fig. 5.2: Evolution of complete travels performance relative to the direct route path system performance. The evolution is according to different combinations of angle and depth view.

5.3 Evaluation limitations

The previous series of simulations provided valuable insights into the various path system patterns that can emerge under specific configurations. These results allowed us to evaluate the strengths and weaknesses of different agent morphologies in generating efficient and functional solutions. However, it is important to note that the presented graphs and analyses are specifically based on an equilateral triangular configuration of hotspots.

The spatial distribution and number of hotspots play a crucial role in shaping the performance of the path system. A uniform and symmetrical arrangement, such as the equilateral triangle configuration, naturally facilitates the formation of optimized patterns like Steiner trees, as agents can easily identify and share common paths. In contrast, an irregular or asymmetrical distribution of hotspots introduces additional complexity, potentially leading to less predictable patterns and varying degrees of efficiency.

These variations in hotspot distribution and density highlight the adaptability and limitations of different agent configurations. As noticed from previous simulation example, a deadlock in the resulting pattern can occur. Certain agent setups, such as those with a high angle and depth of view, may perform well in symmetrical configurations by efficiently consolidating paths, but their effectiveness may diminish in irregular layouts. Conversely, configurations with limited visibility may excel in environments with clearly defined and localized hotspots but struggle in larger or more dispersed networks.

Future studies should explore the impact of different hotspot arrangements on agent behavior and path system performance.

第6章 Conclusion

In this paper, we presented a multi-agent model that reproduces desire paths phenomenon and slime mold shrinkage and readaptation ability.

After a short summary, we will explain the future works and potential improvements that could be added to complete our work.

6.1 Summary

The presented model provides a useful tool that is able to minimize the total size of the paths system without influencing significantly the number of complete travels in a limited time period. Results confirm the convergence of the model while readapting in the absence of more constraints in the environment simulation. But this constant readaptation in some cases can highlight potential deadlocks in simulation.

The influence of different configurations of agent morphology has been tested and performance graphs have been drawn to find the best agent parameters to get the optimal path system considering the importance of performance we look for.

Moreover, the modularity of the model, like the customization of the bias target selection algorithm and the multi-layered environment data implementation possibility show the usefulness of the model in diverse applications.

6.2 Future work and potential improvements

Building on the findings of this study, several future research can be explored to enhance the model and its application. One potential direction is to integrate real-world data, such as the transportation patterns in Tokyo, to compare to our model accuracy and applicability. By analyzing origin-destination data from train users, it is possible to extract probability distributions for selecting destinations based on the starting point. These probabilities could then be applied to hotspots within the simulation, representing train stations, providing a more realistic simulation of commuter behavior. Comparing the resulting path system generated by the model with the actual railway system would offer valuable insights into the accuracy and efficiency of the simulated network, as well as potential areas for optimization.

However, for better accuracy, an auto-balancing traffic model has to be implemented.

Additionally, further investigation could be conducted by experimenting with different weight functions for the pixels, which would affect how agents evaluate paths and make decisions. This could involve testing the impact of varying the weight functions on the performance of the model, providing a deeper understanding of how different factors influence the overall behavior and efficiency of the path system.

In terms of improving the model's realism, an enhancement could be made to the agent behavior algorithm. Currently, the model assumes a straight-line approach, where agents follow the shortest direct path between origin and destination. An interesting improvement would be to integrate an A* pathfinding algorithm, which would enable agents to take into account not only the distance but also other factors such as obstacles or varying terrain. This would make the agents' behavior more nuanced and closer to real-world pathfinding, where multiple factors influence the choice of route.

Furthermore, incorporating a terrain map could allow agents to adjust their routes based on accessibility and environmental constraints. For example, agents would need to avoid areas that are inaccessible or too costly to traverse, such as mountains or bodies of water. While this improvement would introduce more complexity to the model, it would also create a more sophisticated simulation that better represents real-world challenges in route selection and optimization. However, this would require handling more advanced scenarios where agents must make decisions based on their surroundings and the availability of paths, which would add an extra layer of complexity to the simulation.

Another potential enhancement involves utilizing the agent morphology, which allows changes to the focus distance and angle. By dynamically adjusting these parameters according to the agents' "changing" environment as they move, the model can better reproduce human adaptation behavior. This adaptation mechanism could enable agents to respond more effectively to environmental variations and uncertainties, creating a simulation that more accurately mirrors real-world human decision-making processes.



Fig. 6.1: Example of future application of the model for urban planning.

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